

Chapter 6

Item Response Theory Methods

Item response theory (IRT) methods are used in many testing applications, and the use of IRT has been informed by a variety of general treatments (e.g., Baker and Kim 2004; de Ayala 2009; Hambleton and Swaminathan 1985; Hambleton et al. 1991; Lord 1980; Nering and Ostini 2010; Reckase 2009; van der Linden and Hambleton 1997; Wright and Stone 1979; Yen and Fitzpatrick 2006). Applications of IRT include test development, item banking, differential item functioning, adaptive testing, test equating, and test scaling. A major appeal of IRT is that it provides an integrated psychometric framework for developing and scoring tests. Much of the power of IRT results from the fact that it explicitly models examinee responses at the item level, whereas, for example, the focus of classical test models and strong true score models is on responses at the level of test scores.

Unidimensional IRT models have been developed for tests that are intended to measure a single dimension, and *multidimensional IRT models* have been developed for tests that are intended to measure simultaneously along multiple dimensions. IRT models have been developed for tests whose items are scored dichotomously (0/1) as well as for tests whose items are scored polytomously (e.g., a short answer test in which examinees can earn a score of 0, 1, or 2 on each item). See Thissen and Steinberg (1986) for a taxonomy of IRT models.

Many testing programs use unidimensional IRT models to assemble tests. In these testing programs, the use of IRT equating methods often seems natural. Also, IRT methods can be used for equating in some situations in which traditional methods typically are not used, such as equating to an item pool. Thus, IRT methods are an important component of equating methodology. However, IRT models gain their flexibility by making strong statistical assumptions, which likely do not hold precisely in real testing situations. For this reason, studying the robustness of the models to violations of the assumptions, as well as studying the fit of the IRT model, is a crucial aspect of IRT applications. See Hambleton and Swaminathan (1985) and Hambleton et al. (1991) for general discussions of testing model fit, and see von Davier and Wilson (2007) for a detailed discussion of the assumptions made in IRT equating

with the common item nonequivalent groups design along with an example of how to test these assumptions.

The initial focus of this chapter is on equating scores on test forms that contain dichotomously (0/1) items using the unidimensional IRT model referred to as the *three-parameter logistic model* (Lord 1980). This model, which is described more fully later in this chapter, is the most general unidimensional model for dichotomously scored test items that is in widespread use. The *Rasch model* (Rasch 1960; Wright and Stone 1979) also is discussed briefly. In this chapter, after an introduction to IRT, methods of transforming IRT scales are discussed. Then IRT true score equating and IRT observed score equating are treated. The methods are illustrated using the same data that were used in Chaps. 4 and 5. Equating using IRT-based item pools also is discussed. Equating with polytomous IRT models is considered near the end of this chapter. Issues in equating computer administered and computer adaptive tests are considered in Chap. 8.

As is described more fully later in this chapter, equating using IRT typically is a three-step process. First, item parameters are estimated using computer software. Second, parameter estimates are scaled to a base IRT scale using a linear transformation. Third, if number-correct scoring is used, number-correct scores on the new form are converted to the number-correct scale on an old form and then to scale scores.

6.1 Some Necessary IRT Concepts

A description of some necessary concepts in IRT for tests consisting of dichotomously scored items is presented here to provide a basis for understanding unidimensional IRT equating of dichotomously scored tests. References cited earlier provide a much more complete presentation of IRT. Instructional modules on IRT by Harris (1989) and on IRT equating by Cook and Eignor (1991) can be used as supplements to the material presented here.

6.1.1 Unidimensionality and Local Independence Assumptions

Unidimensional item response theory (IRT) models for dichotomously (0/1) scored tests assume that *examinee ability* is described by a single latent variable, referred to as θ , defined so that $-\infty < \theta < \infty$. The use of a single latent variable implies that the construct being measured by the test is *unidimensional*. In practical terms, the unidimensionality assumption in IRT requires that tests measure only one ability. For example, a mathematics test that contains some items that are strictly computational and other items that involve verbal material likely is not unidimensional.

The *item characteristic curve* for each item relates the probability of correctly answering the item to examinee ability. The item characteristic curve for item j is

symbolized by $p_j(\theta)$, which represents the probability of correctly answering item j for examinees with ability θ . For example, if 50% of the examinees with ability $\theta = 1.5$ can be expected to answer item 1 correctly, then the probability can be symbolized as $p_1(\theta = 1.5) = .5$. Note that p_j is written as a function of the variable θ . IRT models typically assume a specified functional form for the item characteristic curve, which is what distinguishes IRT models from one another.

An assumption of *local independence* is made in applying IRT models. Local independence means that, after taking into account examinee ability, examinee responses to the items are statistically independent. Under local independence, the probability that examinees of ability θ correctly answer *both* item 1 *and* item 2 equals the product of the probability of correctly answering item 1 and the probability of correctly answering item 2. For example, if examinees of ability $\theta = 1.5$ have a .5 probability of answering item 1 correctly and a .6 probability of answering item 2 correctly, for such examinees the probability of correctly answering *both* items correctly under local independence is $.30 = .50(.60)$.

The local independence assumption implies that there are no dependencies among examinee responses to items other than those that are attributable to latent ability. One example where local independence likely would not hold is when tests are composed of sets of items that are based on common stimuli, such as reading passages or charts. In this case, local independence probably would be violated because examinee responses to items associated with one stimulus are likely to be more related to one another than examinee responses to items associated with another stimulus.

Although the IRT unidimensionality and local independence assumptions might not hold strictly, they might hold closely enough for IRT to be used advantageously in many practical situations. In using IRT equating, it is important to choose an equating design that minimizes the effects of violations of model assumptions.

6.1.2 IRT Models

Various IRT models are in use that differ in the functional form of the item characteristic curve. Among unidimensional models, the three-parameter logistic model is the most general of the forms in widespread use. In this model, the functional form for an item characteristic curve is characterized by three item parameters. Under the three-parameter logistic model, the probability that persons of ability equal to the ability of person i correctly answer item j is defined as

$$p_{ij} = p_{ij}(\theta_i; a_j, b_j, c_j) = c_j + (1 - c_j) \frac{\exp[Da_j(\theta_i - b_j)]}{1 + \exp[Da_j(\theta_i - b_j)]}. \quad (6.1)$$

In this equation, θ_i is the ability parameter for person i . Ability, θ , is defined over the range $-\infty < \theta < \infty$ and often is scaled to be normally distributed with a mean of 0 and standard deviation of 1. In this case, nearly all of the persons have θ values in the range -3 to $+3$. The expression “exp” in Eq. (6.1) stands for the natural

exponential function. That is, the quantity in brackets after $\exp$ is the exponent of $e = 2.71828 \dots$. The constant D typically is set to 1.7 so that the logistic item response curve and the normal ogive differ by no more than .01 for all values of θ .

The item parameters a_j , b_j , and c_j are associated with item j . The meanings of these parameters are illustrated in the portion of Table 6.1 labeled “Item Parameters” and in Fig. 6.1. For now, consider only the item parameters for the three items listed below the labeled portion “Scale I” on the left-hand side of the table. Also ignore the I subscript for the present.

The item parameter c_j is the *lower asymptote or pseudo-chance level parameter* for item j . The parameter c_j represents the probability that an examinee with very low ability (actually, $\theta = -\infty$) correctly answers the item. For example, for low ability examinees, the curve for item 3 in Fig. 6.1 appears to be leveling off (have a lower asymptote) at a probability of .18, which corresponds to the c -parameter for this item listed in Table 6.1. If the horizontal axis in Fig. 6.1 were extended beyond $\theta = -3$, items 1 and 2 would appear to have the lower asymptotes of .10 and .17 shown in Table 6.1. The c -parameter for an item must be in the range 0 to 1. Typically, the c -parameter for an item is somewhere in the range of 0 to the probability of correctly answering an item by random guessing (1 divided by the number of options).

The item parameter b_j is referred to as the *difficulty or location parameter* for item j . The logistic curve has an inflexion point at $\theta = b$. When $c = 0$, b is the level of ability where the probability of a correct answer is .5. Otherwise, b is the ability level where the probability of a correct response is halfway between c and 1.0. The inflexion point of each curve is indicated by the circular symbol on each item characteristic curve in Fig. 6.1. Typically, b is in the range -3 to $+3$. Higher values of b are associated with more difficult items. As an illustration, item 3 has the highest b -parameter in Table 6.1. Of the three items in Fig. 6.1, the item characteristic curve for item 3 tends to be shifted the farthest to the right.

The item parameter a_j is referred to as the *discrimination parameter* for item j . The a -parameter is proportional to the slope of the item characteristic curve at the inflexion point. As can be seen in Table 6.1, item 3 has the highest a -parameter (1.7) and item 3 also has the steepest item characteristic curve in Fig. 6.1.

The abilities for two persons are shown in the middle of Table 6.1 under the heading “Person Abilities.” The probabilities of correctly answering each of the three items for examinees of ability $\theta = -2.00$ and $\theta = 1.00$ are shown at the bottom of Table 6.1 under the heading “Probability of Correctly Answering Items.” For example, the probability of person $i = 1$ with ability $\theta_{1i} = -2.00$ correctly answering the first item can be calculated as follows using Eq. (6.1):

$$p_{ij} = .10 + (1 - .10) \frac{\exp\{1.7(1.30)[-2.00 - (-1.30)]\}}{1 + \exp\{1.7(1.30)[-2.00 - (-1.30)]\}} = .26.$$

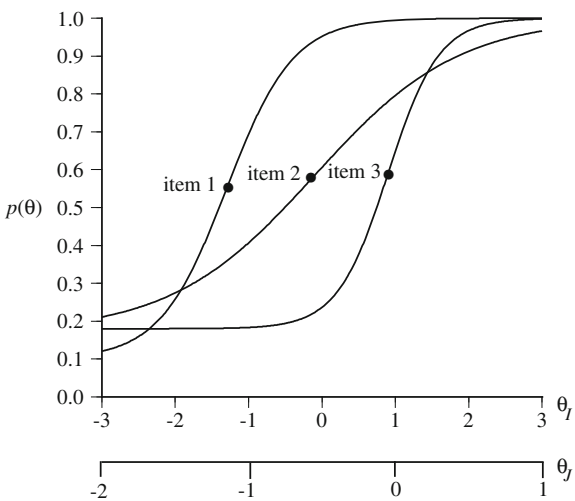
The reader should verify the computation of the other probabilities by substituting the abilities and item parameters into Eq. (6.1).

Various simplifications of the three-parameter logistic model have been used. One variation can be obtained by setting c_j equal to a constant other than 0. The

Table 6.1 Item and person parameters on two scales for a hypothetical test

Scale <i>I</i>				Scale <i>J</i>		
Item parameters						
Item	a_{Ij}	b_{Ij}	c_{Ij}	a_{Jj}	b_{Jj}	c_{Jj}
$j = 1$	1.30	−1.30	.10	2.60	−1.15	.10
$j = 2$	.60	−.10	.17	1.20	−.55	.17
$j = 3$	1.70	.90	.18	3.40	−.05	.18
Person abilities						
Person	θ_{Ii}			θ_{Ji}		
$i = 1$	−2.00			−1.50		
$i = 2$	1.00			.00		
Scale transformation constants						
$A = .5$		$B = −.5$				
Probability of correctly answering items						
$p_{ij}(\theta_{Ii}; a_{Ij}, b_{Ij}, c_{Ij})$				$p_{ij}(\theta_{Ji}; a_{Jj}, b_{Jj}, c_{Jj})$		
Person				Person		
Item	$i = 1$	$i = 2$		$i = 1$	$i = 2$	
$j = 1$	.26	.99		.26	.99	
$j = 2$	.27	.80		.27	.80	
$j = 3$	.18	.65		.18	.65	

Fig. 6.1 Hypothetical example of scale transformations



two-parameter logistic model is obtained from Eq.(6.1) by setting c_j equal to 0. This model does not explicitly accommodate examinee guessing. The Rasch model is obtained from Eq.(6.1) by setting c_j equal to 0, a_j equal to 1, and D equal to 1. The Rasch model, therefore, requires all items to be equally discriminating, and it does

not explicitly accommodate guessing. Other models exist that use a normal ogive to model p_{ij} .

The three-parameter logistic model is the only one of the three models presented that explicitly accommodates items which vary in difficulty, which vary in discrimination, and for which there is a nonzero probability of obtaining the correct answer by guessing. Because of its generality, the three-parameter model is the focus of this chapter. However, the assumed form of the relationship between ability and the probability of a correct response (e.g., the three-parameter logistic curve) is chosen primarily for reasons of mathematical tractability. No reason exists for this relationship to hold, precisely, for actual test items.

6.1.3 IRT Parameter Estimation

IRT item and ability parameters need to be estimated when using IRT methods in practice. Two general approaches to estimating item and ability parameters are *joint maximum likelihood* and *marginal maximum likelihood*. These estimation procedures are described in detail by Baker and Kim (2004) and de Ayala (2009) and are only briefly summarized here.

In joint maximum likelihood, preliminary ability estimates along with examinee item responses are used to estimate item parameters by maximum likelihood procedures. The estimated item parameters are then used to update the ability estimates by maximum likelihood procedures. The updated ability estimates are used to update the estimates of the item parameters, and this type of back-and-forth procedure is repeated until the parameter estimates stabilize. The LOGIST computer software (Wingersky et al. 1982) uses joint maximum likelihood methods with the three-parameter logistic model. LOGIST is not used very much because parameter estimation for the three-parameter logistic model appears to be more stable, and is on a firmer statistical foundation, with marginal maximum likelihood methods. Parameter estimation for the Rasch model often is conducted using joint maximum likelihood procedures using computer software such as WINSTEPS (Linacre 2001).

Marginal maximum likelihood begins by specifying a prior probability distribution for ability (often standard normal) in the population of examinees. Item parameters are estimated assuming this prior distribution of ability. The prior distribution of ability often is updated during the estimation process. The outcome from applying marginal maximum likelihood methods is a set of item parameter estimates for each of the items and a posterior distribution of examinee ability.

Examinee ability parameter estimates are not provided by the marginal maximum likelihood method. Ability parameters can be estimated for each examinee from the examinee's item responses along with the item parameter estimates and the posterior ability distribution that result from application of the marginal maximum likelihood method. Computer software such as BILOG-MG (Zimowski et al. 2003), ICL (Hanson 2002), MULTILOG (Thissen et al. 2003), and PARSCALE (Muraki and Bock 2003) can be used to implement the marginal maximum likelihood method.

Unless the user chooses otherwise, ability parameters are scaled to have a mean of approximately 0 and standard deviation of approximately 1 with these software packages.

One important characteristic of ability estimation for the three-parameter logistic model is that the ability estimates depend on the pattern of item responses, rather than just on the number of items an examinee answers correctly. That is, examinees who earn the same number-correct score would likely earn different estimated θ 's if some of the items that they correctly answered were different. The use of such *pattern scoring* in IRT increases the precision of the IRT ability estimates over using the number-correct score if the IRT model holds. However, for many practical applications, including equating, number-correct scoring often is used. Different IRT ability parameter estimates are discussed in Chap. 9.

6.2 Transformations of IRT Scales

When conducting equating with nonequivalent groups, the parameters from different test forms need to be on the same IRT scale. However, the parameter estimates that result from IRT parameter estimation procedures are often on different IRT scales. For example, assume that the parameters for the IRT model are estimated for Form X based on a sample of examinees from Population 1 and separately for Form Y based on a sample of examinees from Population 2, where the two populations are not equivalent. As was already indicated, computer software often defines the θ -scale as having a mean of 0 and a standard deviation of 1 for the set of data being analyzed. In this case, the abilities for each group would be scaled to have a mean of 0 and a standard deviation of 1, even though the groups differed in ability. Thus, a transformation of IRT scales is needed.

As is demonstrated later in this section, if an IRT model fits a set of data, then any linear transformation of the θ -scale also fits the set of data, provided that the item parameters also are transformed. When the IRT model holds, the parameter estimates from different computer runs are on linearly related θ -scales. Thus, a linear equation can be used to convert IRT parameter estimates to the same scale. After conversion, the means and standard deviations of the abilities for the two groups on the common scale would be expected to differ. The resulting transformed parameter estimates, which sometimes are referred to as being *calibrated*, then can be used to establish score equivalents between number-correct scores on Form X and Form Y, and then to scale scores.

6.2.1 Transformation Equations

Define Scale I and Scale J as three-parameter logistic IRT scales that differ by a linear transformation. Then the θ -values for the two scales are related as follows:

$$\theta_{Ji} = A\theta_{Ii} + B, \quad (6.2)$$

where A and B are constants in the linear equation and θ_{Ji} and θ_{Ii} are values of θ for individual i on Scale J and Scale I . The item parameters on the two scales are related as follows:

$$a_{Jj} = \frac{a_{Ij}}{A}, \quad (6.3)$$

$$b_{Jj} = Ab_{Ij} + B, \quad (6.4)$$

and

$$c_{Jj} = c_{Ij}, \quad (6.5)$$

where a_{Jj} , b_{Jj} , and c_{Jj} are the item parameters for item j on Scale J and a_{Ij} , b_{Ij} , and c_{Ij} are the item parameters for item j on Scale I . The lower asymptote parameter is independent of the scale transformation, as is indicated by Eq. (6.5).

6.2.2 Demonstrating the Appropriateness of Scale Transformations

To demonstrate that there is an A and a B which result in the scale transformation that correctly transforms parameters from Scale I to Scale J , note that the right-hand side of Eq. (6.1) for Scale J equals

$$c_{Jj} + (1 - c_{Jj}) \frac{\exp[Da_{Jj}(\theta_{Ji} - b_{Jj})]}{1 + \exp[Da_{Jj}(\theta_{Ji} - b_{Jj})]}.$$

Now replace θ_{Ji} , a_{Jj} , b_{Jj} , c_{Jj} with the expressions from Eqs. (6.2)–(6.5) as follows:

$$\begin{aligned} c_{Ij} + (1 - c_{Ij}) \frac{\exp \left\{ D \frac{a_{Ij}}{A} [A\theta_{Ii} + B - (Ab_{Ij} + B)] \right\}}{1 + \exp \left\{ D \frac{a_{Ij}}{A} [A\theta_{Ii} + B - (Ab_{Ij} + B)] \right\}} \\ = c_{Ij} + (1 - c_{Ij}) \frac{\exp[Da_{Ij}(\theta_{Ii} - b_{Ij})]}{1 + \exp[Da_{Ij}(\theta_{Ii} - b_{Ij})]}. \end{aligned}$$

This resulting expression is the right-hand portion of Eq. (6.1) for Scale I , which demonstrates that A and B in Eqs. (6.2)–(6.5) provide the scale transformation.

6.2.3 Expressing A and B Constants

One way to express the constants A and B is as follows. For any two individuals, i and i^* , or any two items, j and j^* , A and B in Eqs. (6.2)–(6.5) can be expressed as

$$A = \frac{\theta_{ji} - \theta_{ji^*}}{\theta_{li} - \theta_{li^*}} = \frac{b_{jj} - b_{jj^*}}{b_{lj} - b_{lj^*}} = \frac{a_{lj}}{a_{jj}} \quad (6.6)$$

and

$$B = b_{jj} - Ab_{lj} = \theta_{ji} - A\theta_{li}. \quad (6.7)$$

To illustrate these equalities, refer back to Table 6.1 and Fig. 6.1 for a hypothetical example of scale transformations. Parameters for three items are presented in the portion of Table 6.1 labeled “Item Parameters.” Parameters for these items are given for Scale I and for Scale J . The item characteristic curves for these three items are presented in Fig. 6.1. Note that horizontal scales are presented in this figure for Scale I and Scale J , and these are labeled θ_I and θ_J . As is evident from this figure, the item characteristic curves are the same shape on either scale. To calculate A from Eq. (6.6) using the difficulty parameters for items 1 and 2 ($j = 1$ and $j^* = 2$), take

$$A = \frac{(-1.15) - (-.55)}{(-1.30) - (-.10)} = \frac{-.6}{-1.2} = .5.$$

Alternatively, using the slope parameters for item 1,

$$A = \frac{1.3}{2.6} = .5.$$

Using Eq. (6.7) with the difficulty parameters for item 1,

$$B = (-1.15) - (.5)(-1.30) = -.5.$$

These values agree with those in the section labeled “Scale Transformation Constants” in Table 6.1. Equations (6.6) and (6.7) also can be used to calculate A and B using the θ -values for Persons 1 and 2. These A and B values can be used to transform parameters from Scale I to Scale J using Eqs. (6.2)–(6.5). For example, to transform the ability of Person 1 from Scale I to Scale J using Eq. (6.2), take

$$\theta_{J1} = A\theta_{I1} + B = .5(-2.00) + (-.5) = -1.5,$$

which is the value for Person 1 shown under “Person Abilities” in Table 6.1. To convert the parameters for item 3 from Scale I to Scale J using Eqs. (6.3)–(6.5), take

$$a_{J3} = \frac{a_{I3}}{A} = \frac{1.7}{.5} = 3.4,$$

$$b_{J3} = Ab_{I3} + B = .5(.90) - .5 = -.05,$$

and

$$c_{J3} = c_{I3} = .18.$$

These values agree with the Scale J values in the portion of Table 6.1 labeled “Item Parameters.”

The p_{ij} values based on Eq. (6.1) are presented in the portion of Table 6.1 labeled “Probability of Correctly Answering Items.” These values can be calculated from the item and person parameters presented in Table 6.1; they are the same for Scales I and J , and the p_{ij} values will be identical for any linearly related scales. This property often is referred to as *indeterminacy of scale location and spread*.

6.2.4 Expressing A and B Constants in Terms of Groups of Items and/or Persons

So far, the relationships between scales have been expressed by two abilities and two items. Often, it is more useful to express the relationships in terms of groups of items or people. From Eqs. (6.6) and (6.7) it follows that (see Exercise 6.3)

$$A = \frac{\sigma(b_J)}{\sigma(b_I)}, \quad (6.8a)$$

$$= \frac{\mu(a_I)}{\mu(a_J)}, \quad (6.8b)$$

$$= \frac{\sigma(\theta_J)}{\sigma(\theta_I)}, \quad (6.8c)$$

$$B = \mu(b_J) - A\mu(b_I), \quad \text{and} \quad (6.9a)$$

$$= \mu(\theta_J) - A\mu(\theta_I). \quad (6.9b)$$

The means $\mu(b_J)$, $\mu(b_I)$, $\mu(a_I)$, and $\mu(a_J)$ in these equations are defined over one or more items with parameters that are expressed on both Scale I and Scale J . The standard deviations $\sigma(b_J)$ and $\sigma(b_I)$ are defined over two or more items with parameters that are expressed on both Scale I and Scale J . The means $\mu(\theta_J)$ and $\mu(\theta_I)$ are defined over one or more examinees with ability parameters that are expressed on both Scale I and Scale J . The standard deviations $\sigma(\theta_J)$ and $\sigma(\theta_I)$ are defined over two or more examinees with parameters that are expressed on both Scale I and Scale J .

To illustrate the use of Eqs. (6.8a), (6.8b), and (6.9a), the following quantities can be calculated for the three items from the example in Table 6.1: $\mu(b_I) = -.1667$, $\sigma(b_I) = .8994$, $\mu(a_I) = 1.2$, $\mu(b_J) = -.5833$, $\sigma(b_J) = .4497$, and $\mu(a_J) = 2.4$. From Eqs. (6.8) and (6.9),

$$A = \frac{\sigma(b_J)}{\sigma(b_I)} = \frac{\mu(a_I)}{\mu(a_J)} = \frac{.4497}{.8994} = \frac{1.2000}{2.4000} = .5000,$$

and

$$B = \mu(b_J) - A\mu(b_I) = -.5833 - .5000(-.1667) = -.5000.$$

Similar calculations can be made using the mean and standard deviations for the two ability scales in Table 6.1.

In equating with nonequivalent groups, parameter estimates for the common items would be available for examinees in the two groups. The parameter estimates on the common items could be used to find the scaling constants by substituting estimates for these parameters in the preceding equations.

Consider a situation in which the mean and standard deviation of the abilities on Scale *I* are known for one group of examinees. Also, the mean and standard deviation of the abilities are known for a different group of examinees on Scale *J*. Is there any way Eqs. (6.8c) and (6.9b) can be used to transform Scale *I* to Scale *J*? No! These equations can be used only if the parameters for the *same* group of examinees are expressed on *both* scales.

Consider a different situation, in which the mean and standard deviation of abilities on Scale *I* are 0 and 1, respectively. For the *same* group of examinees, the mean and standard deviation of abilities are 50 and 10, respectively, on Scale *J*. Can Eqs. (6.8c) and (6.9b) be used to transform parameters from Scale *I* to Scale *J*? Yes. The resulting scaling constants calculated using Eqs. (6.8c) and (6.9b) are as follows:

$$A = \frac{\sigma(\theta_J)}{\sigma(\theta_I)} = \frac{10}{1} = 10 \quad \text{and} \quad B = \mu(\theta_J) - A\mu(\theta_I) = 50 - 10(0) = 50.$$

These equations might be used to transform IRT parameters to a different scale when the means and standard deviations of the abilities are known.

6.3 Transforming IRT Scales When Parameters are Estimated

The estimation of item parameters complicates the problem of transforming IRT scales. The process that needs to be followed depends on the design used for data collection.

6.3.1 Designs

In the *random groups equating design*, the IRT parameters for Form X can be estimated separately from the parameters for Form Y. If the same scaling convention (e.g., mean of 0 and standard deviation of 1) for ability is used in the separate estimations, then the parameter estimates for the two forms are assumed to be on the same scale without further transformation. No further transformation is assumed to be required because the groups are randomly equivalent, and the abilities are scaled to have the same mean and standard deviation in both groups. If, for some reason, different scaling conventions were used for the two forms, then estimates of the mean and standard deviations of the posterior distributions or of the θ -estimates could be used in place of the mean and standard deviations of the θ -parameters in Eqs. (6.8c) and (6.9b).

In the *single group design with counterbalancing*, the parameters for all examinees on both forms can be estimated together. Because the parameters for the two forms are estimated together on the same examinees, the parameter estimates are assumed to be on the same scale. If the parameters for the two forms are estimated separately using the same scaling conventions, the parameter estimates can be assumed to be on the same scale following the logic discussed previously for the random groups design.

In the *common-item nonequivalent groups equating design*, the Form Y item and ability parameters typically are estimated at the time Form Y is first administered. Consequently, only the Form X parameters need to be estimated when Form X is equated to Form Y. Because the examinees who took Form X are not considered to be equivalent to the examinees who took Form Y, parameter estimates for the two estimations are not on the same scale. However, there is a set of items that is common to the two forms. The estimates of the item parameters for these common items can be used to estimate the scale transformation.

As an alternative, the parameters for Form X and Form Y can be estimated together. This type of estimation is often referred to as *concurrent calibration* (Wingersky and Lord 1984). For example, a single run of BILOG-MG (Zimowski et al. 2003) can be conducted using the item level data for Form X and Form Y on the two examinee groups, indicating which items are common to the two forms, and indicating to which group (group taking Form X or group taking Form Y) the examinee belongs. In conducting the parameter estimation it is important to use the multi-group feature of BILOG-MG (MG stands for multigroup) because DeMars (2002) showed that item parameter estimates are biased when using marginal maximum likelihood estimation that does not take into account group differences and examinee groups taking the alternate forms differ in ability.

Another alternative is to fix the item parameters for the common items to those estimated on the old form when calibrating the items on the new form. This process is referred to as *fixed parameter calibration*. When there are substantial differences in ability between the old and new examinee groups and this procedure is used, fixed parameter calibration can lead to biased item parameter estimates. Such bias

occurs because the IRT ability scale typically is defined as having a mean of 0 and standard deviation of 1 for the old group of examinees as well as for the new group of examinees. This issue was identified and shown to lead to bias by Paek and Young (2005) and Kim (2006); in these studies it was found that bias could be reduced using strategies involving multiple runs of IRT estimation software. DeMars and Jurich (2012) described how to avoid such bias in a single run of BILOG-MG. Keller and Keller (2011) and Li et al. (2004) also investigated fixed parameter procedures. Because of the associated complexities, fixed parameter calibration is not considered further in this chapter.

Parameter estimates must be on the same scale to proceed with equating number-correct scores on alternate forms and converting them to scale scores. Methods for equating number-correct scores are described later in this chapter.

6.3.2 Mean/Sigma and Mean/Mean Transformation Methods

The most straightforward way to transform the scales in the common-item nonequivalent groups design is to substitute the means and standard deviations of the item parameter estimates of the common items for the parameters in Eqs. (6.8) and (6.9). After transformation, the item parameter estimates are often referred to as being *calibrated*. One procedure, described by Marco (1977) and referred to here as the *mean/sigma method*, uses the means and standard deviations of the b -parameter estimates from the common items in place of the parameters in Eqs. (6.8a) and (6.9a). In another method, described by Loyd and Hoover (1980) and referred to here as the *mean/mean method*, the mean of the a -parameter estimates for the common items is used in place of the parameters in Eq. (6.8b) to estimate the A -constant. Then, the mean of the b -parameter estimates of the common items is used in place of the parameters in Eq. (6.9a) to estimate the B -constant. The values of A and B then can be substituted into Eqs. (6.2)–(6.5) to obtain the rescaled parameter estimates.

When estimates are used in place of the parameters, or when the IRT model does not hold precisely, the equalities shown in Eqs. (6.8) and (6.9) do not necessarily hold. So, the mean/sigma and the mean/mean methods typically produce different results. One reason that the mean/sigma method is sometimes preferred to the mean/mean method is that estimates of b -parameters are more stable than estimates of the a -parameters. However, Baker and Al-Karni (1991) pointed out that the mean/mean method might be preferable because means are typically more stable than standard deviations, and the mean/mean method uses only means. Empirical research comparing these two methods is inconclusive, so the approach suggested here is to consider both procedures, and compare the raw-to-scale score conversions that result from the application of both methods when equating is conducted.

Mislevy and Bock (1990) recommended a further variation that uses the means of the b -parameters and the geometric means of the a -parameters. Stocking and Lord (1983) also discussed procedures for using robust estimates of the means and standard deviations of estimates of the b -parameters, although they were not satisfied

with the performance of these robust methods. Linn et al. (1981) described a related procedure that weights the item parameter estimates by their standard errors.

6.3.3 Characteristic Curve Transformation Methods

One potential problem with the methods considered so far arises when various combinations of a -, b -, and c -parameter estimates produce almost identical item characteristic curves over the range of ability at which most examinees score. For example, in two estimations an item with very different b -parameter estimates could have very similar item characteristic curves. In this case, the mean/sigma method could be overly influenced by the difference between the b -parameter estimates, even though the item characteristic curves for the items on the two estimations were very similar. This problem arises because the scale conversion methods described so far do not consider all of the item parameter estimates simultaneously.

In response to this problem, Haebara (1980) presented a method that considers all of the item parameters simultaneously, and Stocking and Lord (1983) developed a method similar to Haebara's. Stocking and Lord (1983) referred to both their method and the Haebara method as *characteristic curve methods*. To develop these methods, note that the indeterminacy of scale location and spread property which was described earlier implies that, for ability Scales I and J ,

$$p_{ij}(\theta_{ji}; a_{Jj}, b_{Jj}, c_{Jj}) = p_{ij}\left(A\theta_{ji} + B; \frac{a_{Ij}}{A}, Ab_{Ij} + B, c_{Ij}\right) \quad (6.10)$$

for examinee i and item j . Equation (6.10) states that the probability that examinees of a given ability will answer a particular item correctly is the same regardless of the scale that is used to report the scores.

If estimates are used in place of the parameters in Eq. (6.10), then there is no guarantee that the equality will hold over all items and examinees for any A and B . This lack of equality is exploited by the characteristic curve methods.

Haebara Approach

The function used by Haebara (1980) to express the difference between the item characteristic curves is the sum of the squared difference between the item characteristic curves for each item for examinees of a particular ability. For a given θ_i , the sum, over items, of the squared difference can be displayed as

$$Hdiff(\theta_i) = \sum_{j:V} \left[p_{ij}(\theta_{ji}; \hat{a}_{Jj}, \hat{b}_{Jj}, \hat{c}_{Jj}) - p_{ij}\left(\theta_{ji}; \frac{\hat{a}_{Ij}}{A}, A\hat{b}_{Ij} + B, \hat{c}_{Ij}\right) \right]^2. \quad (6.11)$$

The summation is over the common items ($j:V$). In this equation, the difference between each item characteristic curve on the two scales is squared and summed.

$Hdiff$ then is cumulated over examinees. The estimation process proceeds by finding A and B that minimize the following criterion:

$$Hcrit = \sum_i Hdiff(\theta_i). \quad (6.12)$$

The summation in Eq. (6.12) is over examinees.

Stocking and Lord Approach

In contrast to the Haebara approach, Stocking and Lord (1983) used the square difference of sums, over items,

$$SLdiff(\theta_i) = \left[\sum_{j:V} p_{ij}(\theta_{ji}; \hat{a}_{Jj}, \hat{b}_{Jj}, \hat{c}_{Jj}) - \sum_{j:V} p_{ij} \left(\theta_{ji}; \frac{\hat{a}_{Ij}}{A}, A\hat{b}_{Ij} + B, \hat{c}_{Ij} \right) \right]^2. \quad (6.13)$$

In the Stocking and Lord (1983) approach, the summation is taken over items for each set of parameter estimates before squaring. Note that in IRT, the function

$$\tau(\theta_i) = \sum_j p_{ij}(\theta_i) \quad (6.14)$$

is referred to as the *test characteristic curve*. So, the expression $SLdiff(\theta_i)$ is the squared difference between the test characteristic curves for a given θ_i . In contrast, the expression $Hdiff(\theta_i)$ is the sum of the squared difference between the item characteristic curves for a given θ_i , $SLdiff$ then is cumulated over examinees. The estimation proceeds by finding the combination of A and B that minimizes the following criterion:

$$SLcrit = \sum_i SLdiff(\theta_i). \quad (6.15)$$

The summation in Eq. (6.15) is over examinees. The approach to solving for A and B in Eqs. (6.12) and (6.15) is a computationally intensive iterative approach.

Specifying the Summation Over Examinees

In addition to differences in the function used to express the difference between the characteristic curves described in Eqs. (6.11) and (6.13), these methods differ in how they cumulate the differences between the characteristic curves. Various ways to

specify the examinees have been used in the summations in Eqs. (6.12) and (6.15). Some of these ways are as follows:

1. Sum over estimated abilities of examinees who were administered the old form (Stocking and Lord 1983, used a spaced sample of 200 ability estimates).
2. Sum over estimated abilities of examinees who were administered the new form and sum over estimated abilities of examinees who were administered the old form (Haebara 1980).
3. Sum over estimated abilities that are grouped into intervals and then weight the differences by the proportion of examinees in each interval (Haebara 1980).
4. Sum over a set of equally spaced values of ability (Baker and Al-Karni 1991).
5. If the posterior distribution of ability in the population is estimated and represented by a discrete distribution, which is typically the case when using marginal maximum likelihood estimation, use a weighted summation over the posterior ability distribution for the group taking the new form (Zeng and Kolen 1994).
6. If the posterior distribution of ability in the population is estimated and represented by a discrete distribution, use a weighted summation over the posterior ability distribution for the examinees who were administered the old form and a weighted summation over the posterior ability distribution for the group of examinees who were administered the new form (Kim and Kolen 2007).

A decision needs to be made about which of these options (or others) are used when implementing the characteristic curve procedures. The computer software ST and POLYST that is listed in Appendix B can be used to implement these schemes for summation over examinees; in addition the C computer code described by Brennan et al. (2009, pp. 223–256) can be used. Although research regarding the relative accuracy of linking from these different summation procedures has been inconclusive, Kim and Kolen (2007) recommended that the 6th procedure in the preceding list is preferable because it is symmetric (i.e., the linking function going from Form X to Form Y is the the inverse of the linking function going from Form Y to Form X). In addition, the last method makes use of the estimated posterior distributions which appears to be preferable from a theoretical perspective when using marginal maximum likelihood methods.

Hypothetical Example

A hypothetical example is presented in Table 6.2 that illustrates part of the process of scaling item parameter estimates. Assume that the three items listed are common items in a common-item nonequivalent groups equating design, and that the resulting estimates are on different linearly related ability scales. Estimates of A and B based on these parameter estimates for the mean/sigma and mean/mean methods are presented in the top portion of Table 6.2. The Scale I parameter estimates are converted to Scale J in the middle portion of the table. The results for the two methods differ somewhat. These differences likely would cause some differences in raw-to-scale

Table 6.2 Hypothetical example for characteristic curve methods using estimated parameters

Item	Scale <i>I</i>			Scale <i>J</i>		
	$\hat{a}$	$\hat{b}$	$\hat{c}$	$\hat{a}$	$\hat{b}$	$\hat{c}$
1	.4000	−1.1000	.1000	.5000	−1.5000	.1000
2	1.7000	.9000	.2000	1.6000	.5000	.2000
3	1.2000	2.2000	.1000	1.0000	2.0000	.1000
Mean	1.1000	.6667	.1333	1.0333	.3333	.1333
Sd	.5354	1.3573	.0471	.4497	1.4337	.0471
	Mean/Sigma	Mean/Mean				
<i>A</i>	1.0563	1.0645				
<i>B</i>	−.3709	−.3763				

Item	Scale <i>I</i> Converted to Scale <i>J</i> Using Mean/Sigma Results			Scale <i>I</i> Converted to Scale <i>J</i> Using Mean/Mean Results		
	$\hat{a}$	$\hat{b}$	$\hat{c}$	$\hat{a}$	$\hat{b}$	$\hat{c}$
1	.3787	−1.5328	.1000	.3758	−1.5473	.1000
2	1.6094	.5798	.2000	1.5970	.5817	.2000
3	1.1360	1.9530	.1000	1.1273	1.9656	.1000
Mean	1.0414	.3333	.1333	1.0333	.3333	.1333
Sd	.5069	1.4337	.0471	.5030	1.4449	.0471

Estimated probability of correct response given
 $\theta_i = 0$

Item	Original Scale <i>J</i>	Mean/Sigma	Mean/Mean
1	.8034	.7556	.7559
2	.3634	.3359	.3367
3	.1291	.1202	.1203
sum	1.2959	1.2118	1.2130

score conversions, which could be studied if equating relationships subsequently were estimated.

The probability of a correct response, using Eq. (6.1), is shown in the bottom portion of Table 6.2 for examinees with ability $\theta_i = 0$. In this example, the mean/sigma and mean/mean methods are compared using *Hdiff* and *SLdiff* as criteria. The criteria can be calculated at $\theta_i = 0$ using the estimated probabilities at the bottom of Table 6.2. To calculate *Hdiff*(θ_i) using Eq. (6.11), sum, over items, the squared difference between the estimated probabilities for the original Scale *J* and for the transformed scale that results from the application of one of the methods. For example, for the mean/sigma method,

$$\begin{aligned} Hdiff(\theta_i = 0) &= (.8034 - .7556)^2 + (.3634 - .3359)^2 + (.1291 - .1202)^2 \\ &= .003120. \end{aligned}$$

Similarly, for the mean/mean method,

$$\begin{aligned} Hdiff(\theta_i = 0) &= (.8034 - .7559)^2 + (.3634 - .3367)^2 + (.1291 - .1203)^2 \\ &= .003047. \end{aligned}$$

$Hdiff(\theta_i = 0)$ is smaller for the mean/mean method than it is for the mean/sigma method, indicating that the mean/mean method is somewhat “better” than the mean/sigma method at $\theta_i = 0$ based on $Hdiff(\theta_i)$.

To calculate $SLdiff(\theta_i = 0)$ using Eq.(6.13), the estimated probabilities are summed over items, resulting in the sums listed at the bottom of the table. These sums represent the value of the test characteristic curve at $\theta_i = 0$. For the mean/sigma method,

$$SLdiff(\theta_i = 0) = (1.2959 - 1.2118)^2 = .007073.$$

For the mean/mean method,

$$SLdiff(\theta_i = 0) = (1.2959 - 1.2130)^2 = .006872.$$

$SLdiff(\theta_i = 0)$ is smaller for the mean/mean method than it is for the mean/sigma method, indicating that the mean/mean method is somewhat “better” than the mean/sigma method at $\theta_i = 0$. Thus, the mean/mean method is “better” at $\theta_i = 0$ for both criteria. In using these methods, differences would actually need to be calculated at many values of θ_i .

If the scaling were actually done using the characteristic curve methods, $Hcrit$ and $SLcrit$ would be calculated by summing $Hdiff(\theta_i)$ and $SLdiff(\theta_i)$ over different values of θ_i . Also, the iterative minimization algorithms described by Haebara (1980) and Stocking and Lord (1983) would be used to find the A and B that minimized $Hcrit$ and $SLcrit$. Typically, the mean/mean or mean/sigma method estimates of A and B would be used as starting values in the minimization process.

Comparison of Criteria

Research comparing results for the $Hcrit$ - and $SLcrit$ -based methods have suggested that they produce similar results (Kim and Kolen 2007) or that they slightly favor the $Hcrit$ -based methods (Lee and Ban 2010). Theoretically, the $Hcrit$ methods might be argued to be superior to the $SLcrit$ methods for the following reason: $Hdiff(\theta_i)$ can be 0 only if the item characteristic curves are identical at θ_i , whereas $SLdiff(\theta_i)$ could be 0 even if the item characteristic curves differed. In this sense, $Hdiff(\theta_i)$ might be viewed as being more stringent than $SLdiff(\theta_i)$. On the other hand, it might be argued the $SLcrit$ -based methods are preferable theoretically, because they focus on the difference between test characteristic curves.

One potential limitation of the characteristic curve methods is that they do not explicitly account for the error in estimating item parameters (See Divgi 1985; Kim and Cohen 1992; Ogasawara 2001a; for a method that takes into account such error). The failure to take into account error in estimating item parameters, explicitly, might not be that crucial when the sample size is large and the item characteristic curves are well estimated. However, there are situations in which problems might arise. For example, if considerably larger sample sizes are used to estimate parameters for one form than for another, then ignoring the error in parameter estimates might lead to problems in estimating A and B , and in estimating equating relationships. Empirical research is needed to address this issue. Baker (1996) studied the sampling distribution of A and B for the Stocking and Lord (1983) method. von Davier and von Davier (2011) presented a general statistical modeling approach that provides a framework for many of the scale linking methods.

6.3.4 Comparisons Among Scale Transformation Methods

For dichotomous IRT models, research comparing the characteristic curve methods to the mean/sigma and mean/mean methods has generally found that the characteristic curve methods produce more stable results than the mean/sigma and mean/mean methods (Baker and Al-Karni 1991; Hanson and Béguin 2002; Kim and Cohen 1992; Lee and Ban 2010; Li et al. 2012; Ogasawara 2001b,c). In addition, Ogasawara (2000) found that the mean/mean method was more stable than the mean/sigma method. When Ogasawara (2002) estimated standard errors for item parameters and item characteristic curves, he found that the item characteristic curves could be estimated accurately, even when the item parameters were not estimated very precisely. This finding supports the finding that the test characteristic curve linking methods are more accurate than the mean/mean and mean/sigma methods. Kaskowitz and De Ayala (2001) studied the effects of error in estimating item parameters on the test characteristic curve methods. They found that the methods were robust in the presence of modest amounts of error, and that the methods were more accurate with 15 or 25 common items than with 5 common items.

Using simulation procedures in which the data fit an IRT model, Kim and Cohen (1998) compared scale linking using the Stocking and Lord (1983) test characteristic curve method to concurrent calibration using an earlier version of MULTILOG. They also examined concurrent calibration using an earlier version of BILOG-MG (BILOG 3, Mislevy and Bock 1990) that did not allow for multi-group estimation, even though using concurrent calibration with this program was not strictly appropriate. The simulations were all based on data that fit the three-parameter logistic model. For small numbers of common items, Kim and Cohen (1998) found that concurrent calibration produced less accurate results than did the test characteristic curve method. Also, with small numbers of common items, concurrent calibration with MULTILOG produced less accurate results than BILOG. With large numbers of common items, they found that all of the procedures examined had similar accuracy.

Also using simulation procedures in which the data fit the IRT model, Hanson and Béguin (2002) compared the mean/sigma, mean/mean, Stocking and Lord, Haebara, concurrent calibration methods using BILOG-MG, and concurrent calibration using MULTILOG. In this study, the concurrent calibration procedures produced more accurate results than the test characteristic curve methods. The mean/mean and mean/sigma methods were less accurate than the other methods. Kim and Kolen (2007) also found that concurrent calibration methods produced more accurate scale linking than the characteristic curve methods in a simulation study that used BILOG-MG.

Béguin et al. (2000) and Béguin and Hanson (2001) compared the Stocking and Lord method to concurrent calibration using simulated data that purposefully did not fit the IRT model due to multidimensionality. When groups were nonequivalent and the abilities highly correlated, scaling using the Stocking and Lord method produced more accurate equating than scaling using concurrent calibration. This finding is different from what was found by Hanson and Béguin (2002) and Kim and Kolen (2007) where the data were simulated to fit the IRT model.

As a set, these studies suggest that concurrent calibration with currently available computer software, although more accurate than separate estimation when the data fit the IRT model, might be less robust to violations of the IRT assumptions than separate estimation using test characteristic curve methods to link the scales. One additional benefit of separate estimation is that it facilitates examining item parameter estimates for the common items, as was done in Fig. 6.2. These sorts of plots can be developed only if separate estimation is used, because only one item parameter estimate for each common item is produced in concurrent calibration. In practice, separate estimation using the test characteristic curve methods seems to be safest. Concurrent calibration could be used as an adjunct to the separate estimation method.

If concurrent calibration is not used and software for implementing the test characteristic curve methods is unavailable, then the following process might produce acceptable results. Construct a scatterplot of the IRT a -parameter estimates by plotting the parameter estimates for the common items for both groups. Construct similar scatterplots for the b - and c -parameter estimates. Examine the scatterplots and identify any items that appear to be outliers. The identification of outliers is necessarily a subjective process. Estimate the A - and B -constants with the mean/sigma and mean/mean methods both with and without including the item or items with parameter estimates that appear to be outliers. If the mean/sigma and mean/mean procedures give very different results with the outliers included but similar results with the outliers removed, then consider removing these items. If the results from this procedure are not clear, then the use of the characteristic curve procedure might be the best alternative. Note that even when the characteristic curve procedures are used, it is best to use more than one method, and to examine scatterplots to consider eliminating items with very different parameter estimates. In practice, it might be best to implement each of the methods and evaluate the effects of the differences between the methods on equating relationships and on resulting scale scores. Procedures for choosing among equating results are considered in Chap. 8.

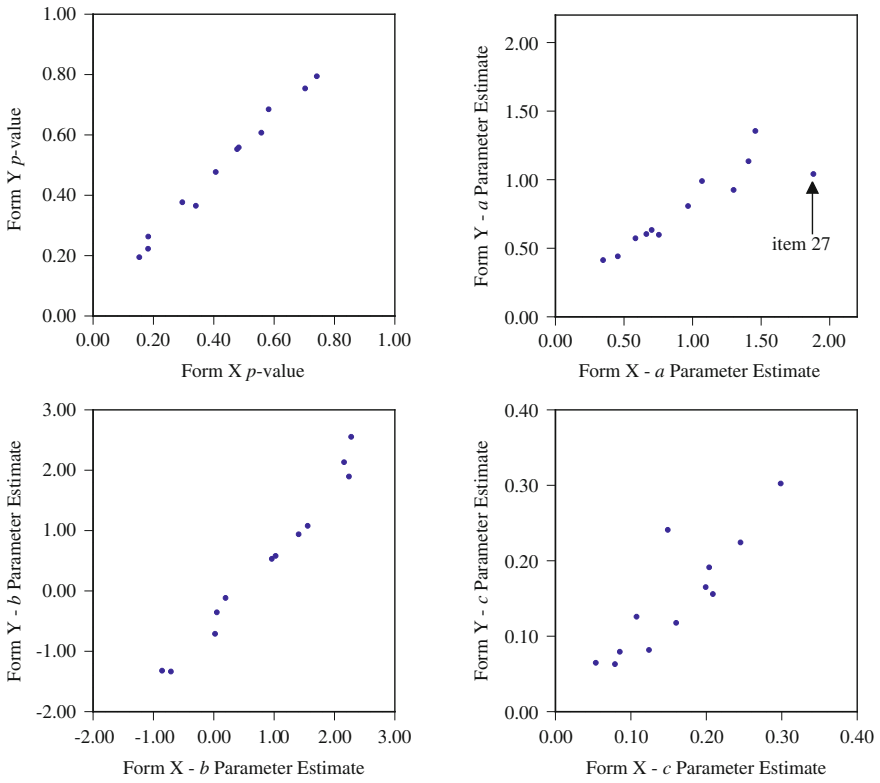


Fig. 6.2 Plots of item parameter estimates on Form X versus Form Y

6.4 Equating and Scoring

When a test is scored using estimated IRT abilities, there is no further need to develop a relationship between scores on Form X and Form Y. Still, the estimated abilities can be converted to scale scores. The ability estimates can be converted so that the reported scores are positive integers, which are presumably easier for examinees to interpret than are scores that may be negative and noninteger, as is the case with estimated IRT abilities. This conversion might involve a linear conversion of estimated abilities, followed by truncating the converted scores so that they are in a specified range of positive numbers, and then rounding the scores to integers for reporting purposes.

However, using estimated IRT abilities results in several practical issues, which might be why they are often not used. One issue is that, to use estimated abilities with the three-parameter logistic model, the whole 0/1 response string, rather than the number-correct score, typically is used to estimate θ . Thus, examinees with the same number-correct score often receive different estimated abilities, which can be

difficult to explain to examinees. In addition, estimates of θ are difficult to compute (they typically cannot be computed by hand). Another concern is that the estimated θ -values with the three-parameter logistic model typically are subject to relatively greater amounts of measurement error for high and low ability examinees than for middle ability examinees. Lord (1980, p. 183) indicated that the measurement error variability for examinees of extreme ability could be 10 or even 100 times that of middle ability examinees, which can create problems in interpreting summary statistics such as means and standard deviations. For these practical reasons, tests often are scored number-correct, even when they are developed and equated using the three-parameter logist IRT model. When number-correct scores are used, an additional step is required in the IRT equating process. The two methods that have been proposed are to equate true scores and to equate observed scores. These procedures are considered next.

6.5 Equating True Scores

After the item parameters are on the same scale, IRT true score equating can be used to relate number-correct scores on Form X and Form Y. In this process, the true score on one form associated with a given θ is considered to be equivalent to the true score on another form associated with that θ .

6.5.1 Test Characteristic Curves

In IRT, the number-correct true score on Form X that is equivalent to θ_i is defined as

$$\tau_X(\theta_i) = \sum_{j:X} p_{ij}(\theta_i; a_j, b_j, c_j), \quad (6.16)$$

where the summation $j:X$ is over items on Form X. The number-correct true score on Form Y that is equivalent to θ_i is defined as

$$\tau_Y(\theta_i) = \sum_{j:Y} p_{ij}(\theta_i; a_j, b_j, c_j), \quad (6.17)$$

where the summation $j:Y$ is over items on Form Y. Equations (6.16) and (6.17) are referred to as *test characteristic curves* for Form X and Form Y. These test characteristic curves relate IRT ability to number-correct true score.

When using the three-parameter logistic model of Eq.(6.1), very low true scores are not attainable with the three-parameter logistic IRT model, because as θ approaches $-\infty$ the probability of correctly answering item j approaches c_j and

not 0. Therefore, true scores on Forms X and Y are associated with a value of θ only over the following ranges (recall that K_X and K_Y are the numbers of items on Form X and Form Y, respectively):

$$\sum_{j:X} c_j < \tau_X < K_X \quad \text{and} \quad \sum_{j:Y} c_j < \tau_Y < K_Y. \quad (6.18)$$

6.5.2 True Score Equating Process

In IRT true score equating, for a given θ_i , true scores $\tau_X(\theta_i)$ and $\tau_Y(\theta_i)$ are considered to be equivalent. The Form Y true score equivalent of a given true score on Form X is

$$\text{irt}_Y(\tau_X) = \tau_Y(\tau_X^{-1}), \quad \sum_{j:X} c_j < \tau_X < K_X, \quad (6.19)$$

where τ_X^{-1} is defined as the θ_i corresponding to true score τ_X . Equation (6.19) implies that true score equating is a three-step process:

1. Specify a true score τ_X on Form X (typically an integer $\sum_{j:X} c_j < \tau_X < K_X$).
2. Find the θ_i that corresponds to that true score (τ_X^{-1}).
3. Find the true score on Form Y, τ_Y , that corresponds to that θ_i .

Form Y equivalents of Form X integer number-correct scores typically are found using these procedures.

Whereas Step 1 and Step 3 are straightforward, Step 2 requires the use of an iterative procedure. For example, suppose that the Form Y equivalent of a Form X score of 5 is to be found. Implementation of Step 2 requires finding the θ_i that results in the right-hand side of Eq. (6.16) equaling 5. Finding this value of θ_i requires the solution of a nonlinear equation using an iterative process. This process is described in the next section.

6.5.3 The Newton-Raphson Method

The Newton-Raphson method is a general method for finding the roots of nonlinear functions. To use this method, begin with a function that is set to 0. Refer to that function as $\text{func}(\theta)$, which is a function of the variable θ . Refer to the first derivative of the function with respect to θ as $\text{func}'(\theta)$. To apply the Newton-Raphson method, an initial value is chosen for θ , which is referred to as θ^- . A new value for θ is calculated as

$$\theta^+ = \theta^- - \frac{\text{func}(\theta)}{\text{func}'(\theta)}. \quad (6.20)$$

Typically, θ^+ will be closer to the root of the equation than θ^- . The new value then is redefined as θ^- , and the process is repeated until θ^+ and θ^- are equal at a specified level of precision or until the value of *func* is close to 0 at a specified level of precision.

When using the Newton-Raphson method, the choice of the initial value is an important consideration, because a poor choice can lead to an erroneous solution. Press et al. (1989) describe modifications to the Newton-Raphson method that are more robust than the Newton-Raphson method to the choice of poor initial values.

Using the Newton-Raphson Method in IRT Equating

To apply this method to IRT true score equating, let τ_X be the true score whose equivalent is to be found. From Eq. (6.16) it follows that θ_i is to be found such that the expression

$$func(\theta_i) = \tau_X - \sum_{j:X} p_{ij}(\theta_i; a_j, b_j, c_j) \quad (6.21)$$

equals 0. The Newton-Raphson method can be employed to find this θ_i using the first derivative of *func*(θ_i) with respect to θ_i , which is

$$func'(\theta_i) = - \sum_{j:X} p'_{ij}(\theta_i; a_j, b_j, c_j) \quad (6.22)$$

where $p'_{ij}(\theta_i; a_j, b_j, c_j)$ is defined as the first derivative of $p_{ij}(\theta_i; a_j, b_j, c_j)$ with respect to θ_i , Lord (1980, p. 61) provided this first derivative:

$$p'_{ij}(\theta_i; a_j, b_j, c_j) = \frac{1.7a_j(1 - p_{ij})(p_{ij} - c_j)}{(1 - c_j)}, \quad (6.23)$$

where $p_{ij} = p_{ij}(\theta_i; a_j, b_j, c_j)$. The resulting expressions for *func*(θ_i) and *func'*(θ_i) are substituted into Eq. (6.20).

A Hypothetical Example

A hypothetical example using this procedure is presented in Table 6.3. In this example, a five-item Form X is to be equated to a five-item Form Y. Parameters (not estimates) are given, and assume that the parameters for the two forms are on the same scale. Table 6.3 shows how to find a Form Y equivalent of a Form X score of 2. The item parameters for Form X are presented in the top portion of the table. To find the Form Y equivalent, the θ_i must be found that corresponds to a Form X score of 2. That is, the θ_i must be found such that, when it is substituted into the right-hand side of Eq. (6.16), it results in a 2 on the left-hand side. The second portion of Table 6.3 illustrates how to find θ_i using the Newton-Raphson method. First, a starting value

Table 6.3 Hypothetical example for IRT true score equating**Form X item parameters**

Item								
Parameter		Item 1	Item 2	Item 3	Item 4	Item 5		
a_j		.60	1.20	1.00	1.40	1.00		
b_j		−1.70	−1.00	.80	1.30	1.40		
c_j		.20	.20	.25	.25	.20		
Solve for $\tau_X = 2$ Using starting value $\theta_i = -2$								
Iteration		Item 1	Item 2	Item 3	Item 4	Item 5	sum	θ_i^+
1	p_{ij}	.5393	.2921	.2564	.2503	.2025	1.5405	−.7941
	p'_{ij}	.1993	.1662	.0107	.0007	.0042	.3811	
2	p_{ij}	.7727	.6828	.2968	.2551	.2187	2.2261	−1.1295
	p'_{ij}	.1660	.3905	.0746	.0121	.0311	.6743	
3	p_{ij}	.7132	.5475	.2772	.2523	.2107	2.0009	−1.1308
	p'_{ij}	.1877	.4010	.0446	.0055	.0180	.6566	
4	p_{ij}	.7130	.5469	.2771	.2523	.2107	2.0000	−1.1308
	p'_{ij}	.1877	.4008	.0445	.0055	.0179	.6564	

Therefore, $\tau_X = 2$ corresponds to $\theta_i = -1.1308$.

Form Y item parameters

Item		Item 1	Item 2	Item 3	Item 4	Item 5		
Parameter		Item 1	Item 2	Item 3	Item 4	Item 5		
a_j		.70	.80	1.30	.90	1.10		
b_j		-1.50	-1.20	.00	1.40	1.50		
c_j		.20	.25	.20	.25	.20		

Form Y true score equivalent of $\theta_i = -1.1308$

	Item 1	Item 2	Item 3	Item 4	Item 5	τ_Y
p_{ij}	.6865	.6426	.2607	.2653	.2058	2.0609

Therefore, $\tau_X = 2$ corresponds to $\tau_Y = 2.0609$.

of $\theta_i^- = -2$ is chosen (this value is an initial guess). Using $\theta_i^- = -2$, the item characteristic curve value from Eq. (6.1) is calculated for each item. For example, the probability of an examinee with an ability of -2 correctly answering item 1 is .5393. The first derivative is also calculated. For example, for the first item, the first derivative of this item evaluated at an ability of -2 can be calculated using Eq. (6.23) as

$$p'_{ij} = \frac{1.7(.60)(1 - .5393)(.5393 - .20)}{(1 - .20)} = .1993,$$

which is also presented in the table.

Next, $func(\theta_i^-)$ from Eq. (6.21) is calculated using 2 for τ_X and the tabled value of 1.5405 as the sum of the item characteristic curves at an ability of -2 . So, $func(\theta_i^-) = 2 - 1.5405$. Then, the negative of the sum of the first derivatives is $func'(\theta_i^-) = -.3811$ from Eq. (6.22). Finally, using Eq. (6.20), the updated ability is

$$\theta_i^+ = \theta_i^- - \frac{func(\theta_i^-)}{func'(\theta_i^-)} = -2 - \frac{2 - 1.5405}{-.3811} = -.7943.$$

The value of $-.7943$ differs in the fourth decimal place from the tabled value because of rounding error; the tabled value is more accurate. The value of $-.7941$ then is used as θ_i^- in the next iteration. The iterations continue until the values of θ stabilize. Note that after the fourth iteration, θ_i^+ equals θ_i^+ after the third iteration, to four decimal places. Also, the sum of the p_{ij} is 2.0000 when $\theta_i = -1.1308$. Thus, a true score of 2 on Form X corresponds to a θ_i of -1.1308 .

The Form Y equivalent of a Form X score of 2 is found next. The Form Y item parameter estimates are needed and are shown in Table 6.3. (Note that the item parameters for Form X and Form Y must be on the same θ -scale.) To find the Form Y equivalent of a Form X score of 2, calculate the value of the item characteristic curve for each Form Y item at $\theta_i = -1.1308$ and sum these values over items. This process is illustrated at the bottom of the table. As shown, a score of 2 on Form X corresponds to a score of 2.0609 on Form Y.

Using the procedures outlined, the reader can verify that a true score of 3 on Form X corresponds to a θ_i of .3655 and a Form Y true score of 3.2586. Also, a true score of 4 on Form X corresponds to a θ_i of 1.3701 and a Form Y true score of 4.0836. Note that a Form X true score of 1 is below the sum of the c -parameters for that form, so the Form Y true score equivalent of a Form X true score of 1 cannot be calculated by the methods described so far.

Sometimes Form Y true score equivalents of all Form X integer scores that are between the sum of the c -parameters and all correct need to be found. The recommended procedure for finding these is to begin with the smallest Form X score that is greater than the sum of the c -parameters. Use a small value of θ as a starting value (e.g., $\theta_i^- = -3$), and then solve for the Form Y true score equivalent. The θ that results from this process can be used as the starting value for solving for the next highest true score. This process continues for all integer true scores on Form X that are below a score of all correct. Sometimes even this procedure can have convergence problems. In this case, the starting values might need to be modified or the modified Newton-Raphson method described by Press et al. (1989) could be tried.

6.5.4 Using True Score Equating with Observed Scores

The true score relationship is appropriate for equating true scores on Form X to true scores on Form Y. However, true scores of examinees are never known, because they are parameters. In practice, the true score equating relationship often is used to convert number-correct observed scores on Form X to number-correct observed scores on Form Y. However, no theoretical reason exists for treating scores in this way. Rather, doing so has been justified in item response theory by showing that the resulting true score conversions are similar to observed score conversions (Lord and Wingersky 1984).

Recall from Eq. (6.18) that the lowest possible true score for the three-parameter IRT model is the sum of the c_j , not 0. Therefore, when using true score equating with observed scores, a procedure is needed for converting Form X scores outside the range of possible true scores on Form X. Lord (1980) and Kolen (1981) presented ad hoc procedures to handle this problem. The Kolen (1981) ad hoc procedure is as follows:

1. Set a score of 0 on Form X equal to a score of 0 on Form Y.
2. Set a score of the sum of the c_j -parameters on Form X equal to the sum of the c_j -parameters on Form Y.
3. Use linear interpolation to find equivalents between these points.
4. Set a score of K_X on Form X equal to a score of K_Y on Form Y.

To formalize this procedure, define τ_X^* as a score outside the range of possible true scores, but within the range of possible observed scores. Equivalents then are defined by the following equation:

$$\begin{aligned} irt_Y(\tau_X^*) &= \frac{\sum_{j:Y} c_j}{\sum_{j:X} c_j} \tau_X^*, & 0 \leq \tau_X^* \leq \sum_{j:X} c_j, \\ &= K_Y, & \tau_X^* = K_X. \end{aligned} \quad (6.24)$$

The use of Kolen's (1981) ad hoc procedure can be illustrated using the hypothetical example presented in Table 6.3. For the item parameters presented, the sum of the c_j -parameters is 1.1 for Form X and 1.1 for Form Y. To apply the procedure to find Form Y equivalents of Form X scores at or below 1.1, take

$$irt_Y(\tau_X^*) = \frac{\sum_{j:Y} c_j}{\sum_{j:X} c_j} \tau_X^* = \frac{1.1}{1.1} \tau_X^* = \tau_X^*.$$

Thus, for example, a score of 1 on Form X is considered to be equivalent to a score of 1 on Form Y. Note that a score of 1 on Form X would have been considered to be equivalent to a score other than 1 on Form Y if the sum of the c_j -parameters was different for the two forms.

In practice, for IRT true score equating, estimates of the item parameters are used to produce an *estimated true score relationship*. Then the estimated true score conversion is applied to the observed scores.

6.6 Equating Observed Scores

Another procedure, *IRT observed score equating*, uses the IRT model to produce an estimated distribution of observed number-correct scores on each form, which then are equated using equipercentile methods. For Form X, the *compound binomial distribution* (see Lord and Wingersky 1984) is used to generate the distribution of

observed number-correct scores for examinees of a given ability. These observed score distributions then are cumulated over a population of examinees to produce a number-correct observed score distribution for Form X. Similar procedures are followed to produce a number-correct observed score distribution for Form Y. The resulting number-correct score distributions then are equated using conventional equipercentile methods. IRT observed score equating requires explicit specification of the distribution of ability in the population of examinees.

Consider a group of examinees all of ability θ_i who have been administered a three-item test with p_{ij} defined by Eq. (6.1). Assuming local independence, the probability that examinees of ability equal to θ_i will incorrectly answer all three items and earn a raw score of 0 is $f(x = 0|\theta_i) = (1 - p_{i1})(1 - p_{i2})(1 - p_{i3})$. To earn a score of 1, an examinee could answer item 1 correctly and items 2 and 3 incorrectly, or the examinee could answer item 2 correctly and items 1 and 3 incorrectly, or the examinee could answer item 3 correctly and items 1 and 2 incorrectly. That is, there are three ways to earn a score of 1 on a three-item test. The probability of earning a 1 is as follows:

$$f(x = 1|\theta_i) = p_{i1}(1 - p_{i2})(1 - p_{i3}) + (1 - p_{i1})p_{i2}(1 - p_{i3}) \\ + (1 - p_{i1})(1 - p_{i2})p_{i3}.$$

The probabilities of correctly answering two and three items can be constructed similarly as follows:

$$f(x = 2|\theta_i) = p_{i1}p_{i2}(1 - p_{i3}) + p_{i1}(1 - p_{i2})p_{i3} + (1 - p_{i1})p_{i2}p_{i3},$$

and

$$f(x = 3|\theta_i) = p_{i1}p_{i2}p_{i3}.$$

Based on the hypothetical example in Table 6.1, for examinees with ability equal to that of Person 1 ($\theta_{I1} = -2.0$),

$$f(x = 0|\theta_1) = (1 - .26)(1 - .27)(1 - .18) = .4430, \\ f(x = 1|\theta_1) = (.26)(1 - .27)(1 - .18) + (1 - .26)(.27)(1 - .18) \\ + (1 - .26)(1 - .27)(.18) \\ = .4167, \\ f(x = 2|\theta_1) = (.26)(.27)(1 - .18) + (.26)(1 - .27)(.18) \\ + (1 - .26)(.27)(.18) \\ = .1277, \\ f(x = 3|\theta_1) = (.26)(.27)(.18) = .0126.$$

Note that these values sum to 1, which is consistent with their being probabilities.

A recursion formula (Lord and Wingersky 1984) can be used to generalize this procedure to more than three items. To implement the recursion formula, define $f_r(x|\theta_i)$ as the distribution of number-correct scores over the first r items for examinees of ability θ_i . Define $f_1(x = 0|\theta_i) = (1 - p_{i1})$ as the probability of earning a score of 0 on the first item and $f_1(x = 1|\theta_i) = p_{i1}$ as the probability of earning a score of 1 on the first item. For $r > 1$, the recursion formula is as follows:

$$\begin{aligned} f_r(x|\theta_i) &= f_{r-1}(x|\theta_i)(1 - p_{ir}), & x = 0 \\ &= f_{r-1}(x|\theta_i)(1 - p_{ir}) + f_{r-1}(x - 1|\theta_i)p_{ir}, & 0 < x < r, \\ &= f_{r-1}(x - 1|\theta_i)p_{ir}, & x = r \end{aligned} \quad (6.25)$$

An example of the use of this recursion formula is presented in Table 6.4. An abbreviated notation is used in this table to simplify the presentation. Specifically, θ_i is dropped and p_r means p_{ir} . To find the distribution for a particular value of r , Eq. (6.25) and Table 6.4 indicate that the distribution for $r - 1$ and the probability of correctly answering item r are needed. Although expressions are only presented up to $r = 4$, the table readily generalizes to higher values of r using the recursion formula. The probabilities listed for the example under $r = 3$ (e.g., .4430, .4167, .1277, and .0126) are identical to results presented earlier.

The procedures presented thus far give the observed score distribution for examinees of a given ability. To find the observed score distribution for examinees of various abilities, the observed score distribution for examinees at each ability is found and then these are accumulated. When the ability distribution is continuous, then

$$f(x) = \int_{\theta} f(x|\theta)\psi(\theta)d\theta, \quad (6.26)$$

where $\psi(\theta)$ is the distribution of θ .

To implement this procedure in practice, some method is needed to perform the integration in Eq. (6.26). Some form of numerical integration is one possibility. When BILOG-MG is used, the distribution of ability typically is characterized by a discrete distribution on a finite number of equally spaced points as a method of approximating the integral. Using this characterization,

$$f(x) = \sum_i f(x|\theta_i)\psi(\theta_i). \quad (6.27)$$

When the distribution of ability is characterized by a finite number of abilities for N examinees, then

$$f(x) = \frac{1}{N} \sum_i f(x|\theta_i). \quad (6.28)$$

This characterization can be used, for example, with a set of abilities that are estimated using BILOG-MG.

Table 6.4 IRT observed score distribution recursion formula example

$r \quad x \quad f_r(x) \text{ for } r \leq 4$		Example (Using Table 6.1 test for persons with $\theta_i = -2$)	
1	0 $f_1(0) = (1 - p_1)$	$= (1 - .26)$	$= .74$
	1 $f_1(1) = p_1$		$= .26$
2	0 $f_2(0) = f_1(0)(1 - p_2)$	$= .74(1 - .27)$	$= .5402$
	1 $f_2(1) = f_1(1)(1 - p_2) + f_1(0)p_2$	$= .26(1 - .27) + .74(.27)$	$= .3896$
	2 $f_2(2) =$	$f_1(1)p_2 =$	$.26(.27) = .0702$
3	0 $f_3(0) = f_2(0)(1 - p_3)$	$= .5402(1 - .18)$	$= .4430$
	1 $f_3(1) = f_2(1)(1 - p_3) + f_2(0)p_3$	$= .3896(1 - .18) + .5402(.18)$	$= .4167$
	2 $f_3(2) = f_2(2)(1 - p_3) + f_2(1)p_3$	$= .0702(1 - .18) + .3896(.18)$	$= .1277$
	3 $f_3(3) =$	$f_2(2)p_3 =$	$.0702(.18) = .0126$
4	0 $f_4(0) = f_3(0)(1 - p_4)$		
	1 $f_4(1) = f_3(1)(1 - p_4) + f_3(0)p_4$		
	2 $f_4(2) = f_3(2)(1 - p_4) + f_3(1)p_4$		
	3 $f_4(3) = f_3(3)(1 - p_4) + f_3(2)p_4$		
	4 $f_4(4) =$	$f_3(3)p_4$	

To conduct observed score equating, observed score distributions are found for Form X and for Form Y. For example, assume that the characterization of the ability distribution associated with Eq. (6.27) is used. The following distributions could be specified using this equation:

1. $f_1(x) = \sum_i f(x|\theta_i)\psi_1(\theta_i)$ is the Form X distribution for Population 1.
2. $f_2(x) = \sum_i f(x|\theta_i)\psi_2(\theta_i)$ is the Form X distribution for Population 2.
3. $g_1(y) = \sum_i g(y|\theta_i)\psi_1(\theta_i)$ is the Form Y distribution for Population 1.
4. $g_2(y) = \sum_i g(y|\theta_i)\psi_2(\theta_i)$ is the Form Y distribution for Population 2.

These quantities then are weighted using synthetic weights described in Chaps. 4 and 5 to obtain the distributions of X and Y in the synthetic population. Conventional equipercentile methods then are used to find score equivalents.

When BILOG-MG is used, the number-correct observed score distributions can be estimated by using the estimated posterior distribution of ability in place of $\psi(\theta_i)$ in Eq. (6.27) along with estimates of $f(x|\theta_i)$ based on substituting estimates for parameters in Eq. (6.25) as suggested by Zeng and Kolen (1995). An alternative is to use the set of estimated abilities in place of the abilities in Eq. (6.28). However, the use of estimates of θ might create systematic distortions in the estimated distributions and lead to inaccurate equating (Han et al. 1997; Lord 1982).

6.7 IRT True Score Versus IRT Observed Score Equating

Compared to IRT observed score equating, IRT true score equating has the advantages of (a) easier computation and (b) a conversion that does not depend on the distribution of ability. However, IRT true score equating has the disadvantage that it equates true scores, which are not available in practice. No justification exists for applying the true score relationship to observed scores. Also, with the three-parameter logistic model, equivalents are undefined at very low scores and at the top number-correct score.

IRT observed score equating has the advantage that it defines the equating relationship for observed scores. Also, assuming reasonable model fit, the distribution of Form X scores converted to the Form Y scale is approximately equal to the distribution of Form Y scores for the synthetic population of examinees. There is no theoretical reason to expect this property to hold for IRT true score equating. Also, using posterior distributions of θ from BILOG-MG, the computational burden of IRT observed score equating is reasonable.

IRT observed score and IRT true score equating methods were found by Kolen (1981) and Han et al. (1997) to produce somewhat different results using the random groups design with achievement tests. However, Lord and Wingersky (1984) concluded that the two methods produce very similar results in a study using the common-item nonequivalent groups design in the SAT.

Larger differences between IRT true and observed score equating might be expected to occur near a number-correct score of all correct and near number-correct scores below the sum of the c -parameter estimates, because these are the regions where IRT true score equating does not produce equivalents. In practice, both methods should be applied with special attention paid to equating results near these regions. Procedures for choosing among the results from equating methods are considered in Chap. 8.

6.8 Illustrative Example

The real data example from Chaps. 4 and 5 is used to illustrate some aspects of IRT equating, using the common-item nonequivalent groups design. Two forms of a 36-item multiple-choice test, Form X and Form Y, are used in this example. Every third item on the test forms is a common item, and the common items are in the same position on each form. Thus, items 3, 6, 9, . . . , 36 on each form represent the 12 common items. Form X was administered to 1,655 examinees and Form Y to 1,638 examinees. As was indicated in Chaps. 4 and 5, the examinees who were administered Form X had a number-correct score mean of 5.11 and a standard deviation of 2.38 on the common items. The examinees who were administered Form Y had a number-correct score mean of 5.87 and a standard deviation of 2.45 on the common items.

Thus, on the common items, the group taking Form Y was higher achieving than the group taking Form X.

6.8.1 Item Parameter Estimation and Scaling

Item parameters were estimated using an earlier version of BILOG-MG (Bilog 3, Mislevy and Bock 1990) separately for each form. (Default parameter settings were used, except for the FLOAT option.) The parameter estimates are given in Table 6.5. The proportion of examinees correctly answering each item (p -value) is also presented.

The Form X item parameter estimates need to be rescaled. The computer software ST that is described in Appendix B was used to conduct the scaling. The common items are tabulated separately in the upper portion of Table 6.6. Because the items appeared in identical positions in the two forms, item 3 on Form X is the same as item 3 on Form Y, and so forth.

The parameter estimates for the common items are plotted in Fig. 6.2 to look for outliers—items with estimates that do not appear to lie on a straight line. In this figure, one item appears to be an outlier for the a -parameter estimate. This item, which is item 27, has a -parameter estimates of 1.8826 on Form X and 1.0417 on Form Y. Because item 27 appears to function differently in the two forms, this item might need to be eliminated from the common-item set. (The c -parameter estimates for item 21 might also be judged to be an outlier, so that item 21 could be considered for elimination as well. This item was not considered for elimination in the present example because it does not seem to be as clearly an outlier as item 27.) Removal of items that appear to be outliers is clearly a judgmental process.

The mean and standard deviation of the item parameter estimates for the common items are shown in Table 6.6. These means and standard deviations were used to estimate the A - and B -constants for transforming the θ -scale of Form X to the θ -scale of Form Y using the mean/mean and mean/sigma methods. For example, using Eqs. (6.8a) and (6.9a) for the mean/sigma method,

$$A = \frac{1.2458}{1.0658} = 1.1689 \quad \text{and} \quad B = .4900 - (1.1689).8602 = -.5155.$$

The B -value differs from the tabled value in the fourth decimal place because of rounding error; the tabled values are more accurate. The A - and B -constants for the Stocking and Lord and Haebara methods that are shown also were calculated using the ST computer software.

Because item 27 appeared to be an outlier, the A - and B -constants were estimated again, eliminating item 27. The means and standard deviations after eliminating this item are shown in Table 6.6 as are the new A - and B -constants. Eliminating item 27 results in the estimates of the A - and B -constants for mean/sigma and mean/mean methods being closer to one another than when item 27 is included. The A - and B -

Table 6.5 Item parameter estimates for common-item equating

Item	Form X				Form Y			
	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$
1	.8440	.5496	−1.7960	.1751	.8527	.8704	−1.4507	.1576
2	.6669	.7891	−.4796	.1165	.6161	.4628	−.4070	.1094
3	.7025	.4551	−.7101	.2087	.7543	.4416	−1.3349	.1559
4	.5405	1.4443	.4833	.2826	.7145	.5448	−.9017	.1381
5	.6723	.9740	−.1680	.2625	.8295	.6200	−1.4865	.2114
6	.7412	.5839	−.8567	.2038	.7946	.5730	−1.3210	.1913
7	.5895	.8604	.4546	.3224	.6351	1.1752	.0691	.2947
8	.6475	1.1445	−.1301	.2209	.6094	.4450	.2324	.2723
9	.5816	.7544	.0212	.1600	.6852	.5987	−.7098	.1177
10	.5296	.9170	1.0139	.3648	.6644	.8479	−.4253	.1445
11	.4825	.9592	.7218	.2399	.7439	1.0320	−.8184	.0936
12	.5574	.6633	.0506	.1240	.6076	.6041	−.3539	.0818
13	.5411	1.2324	.4167	.2535	.5685	.8297	−.0191	.1283
14	.4051	1.0492	.7882	.1569	.6094	.7252	−3155	.0854
15	.4770	1.0690	.9610	.2986	.5532	.9902	.5320	.3024
16	.5139	.9193	.6099	.2521	.5092	.7749	.5394	.2179
17	.5175	.8935	.5128	.2273	.4786	.5942	.8987	.2299
18	.4825	.9672	.1950	.0535	.5587	.8081	−.1156	.0648
19	.4909	.6562	.3953	.1201	.6265	.9640	−.1948	.1633
20	.4081	1.0556	.9481	.2036	.4908	.7836	.3506	.1299
21	.3404	.3479	2.2768	.1489	.3655	.4140	2.5538	.2410
22	.4299	.8432	1.0601	.2332	.5905	.7618	−.1581	.1137
23	.3839	1.1142	.5826	.0644	.5092	1.1959	.5056	.2397
24	.4063	1.4579	1.0241	.2453	.4774	1.3554	.5811	.2243
25	.3706	.5137	1.3790	.1427	.4976	1.1869	.6229	.2577
26	.3077	.9194	1.0782	.0879	.5055	1.0296	.3898	.1856
27	.2956	1.8811	1.4062	.1992	.3771	1.0417	.9392	.1651
28	.2612	1.5045	1.5093	.1642	.3851	1.2055	1.1350	.2323
29	.2727	.9664	1.5443	.1431	.3894	.9697	.6976	.1070
30	.1820	.7020	2.2401	.0853	.2231	.6336	1.8960	.0794
31	.3059	1.2651	1.8759	.2443	.3166	1.0822	1.3864	.1855
32	.2146	.8567	1.7140	.0865	.3356	1.0195	.9197	.1027
33	.1826	1.4080	1.5556	.0789	.2634	1.1347	1.0790	.0630
34	.1814	.5808	3.4728	.1399	.1760	1.1948	1.8411	.0999
35	.1288	.9257	3.1202	.1090	.1424	1.1961	2.0297	.0832
36	.1530	1.2993	2.1589	.1075	.1950	.9255	2.1337	.1259

Note Common-item numbers and parameter estimates are in **boldface** type

constants for the Stocking and Lord and Haebara methods are less affected by eliminating item 27 than are the constants for the mean/sigma and mean/mean methods. In the present example, the scalings based on removing item 27 only are considered for ease of exposition. In practice, however, equating based on scalings with item 27 removed and included could be conducted and the results of the equating compared.

Table 6.6 Common-item parameter estimates and scaling constants

Item	Form X				Form Y			
	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$
3	.7025	.4551	−.7101	.2087	.7543	.4416	−1.3349	.1559
6	.7412	.5839	−.8567	.2038	.7946	.5730	−1.3210	.1913
9	.5816	.7544	.0212	.1600	.6852	.5987	−.7098	.1177
12	.5574	.6633	.0506	.1240	.6076	.6041	−.3539	.0818
15	.4770	1.0690	.9610	.2986	.5532	.9902	.5320	.3024
18	.4825	.9672	.1950	.0535	.5587	.8081	−.1156	.0648
21	.3404	.3479	2.2768	.1489	.3655	.4140	2.5538	.2410
24	.4063	1.4579	1.0241	.2453	.4774	1.3554	.5811	.2243
27	.2956	1.8826	1.4062	.1992	.3771	1.0417	.9392	.1651
30	.1820	.7020	2.2401	.0853	.2231	.6336	1.8960	.0794
33	.1826	1.4080	1.5556	.0789	.2634	1.1347	1.0790	.0630
36	.1530	1.2993	2.1589	.1075	.1950	.9255	2.1337	.1259
$\hat{\mu}$	.4252	.9657	.8602	.1595	.4879	.7934	.4900	.1510
$\hat{\sigma}$	.1917	.4464	1.0658	.0707	.1960	.2837	1.2458	.0736
	Mean/ Sigma	Mean/ Mean	Stocking −Lord	Haebara				
A =	1.1689	1.2173	1.0946	1.0678				
B =	−.5156	−.5572	−.4978	−.4713				
Eliminating Item #27								
$\hat{\mu}$	.4370	.8825	.8106	.1559	.4980	.7708	.4491	.1498
$\hat{\sigma}$	.1961	.3665	1.0999	.0728	.2019	.2858	1.2935	.0768
	Mean/ Sigma	Mean/ Mean	Stocking −Lord	Haebara				
A =	1.1761	1.1449	1.0861	1.0638				
B =	−.5042	−.4790	−.4733	−.4540				

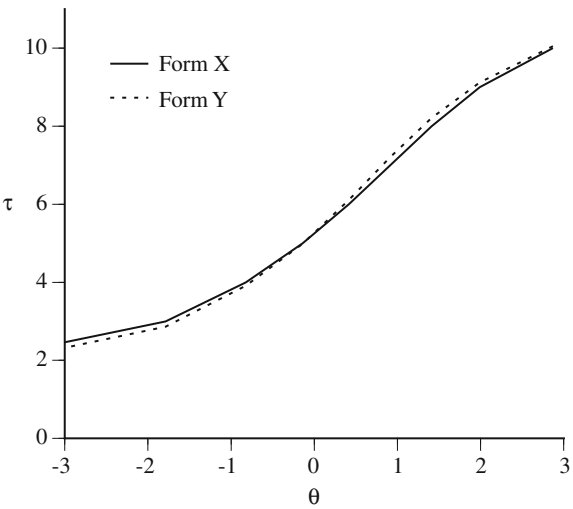
The rescaled Form X item parameter estimates for the common items are shown in Table 6.7. So that all of the computations in this example could be done by hand, the mean/sigma method was used, excluding item 27. Because the Form Y item parameter estimates are not being transformed, they are identical to those in Table 6.6. To verify the tabled Form X *b*-parameter estimate for item 3, take $1.1761(-.7101) - .5042 = -1.3393$, which differs in the fourth decimal place because of rounding. To find the tabled Form X *a*-parameter estimate for this item, take $.4551/1.1761 = .3870$.

The means and standard deviations of the rescaled parameter estimates for the common items are shown at the bottom of Table 6.7. Because the mean/sigma method was used, the mean and standard deviation of the rescaled *b*-parameter estimates for Form X are equal to those for Form Y. Note, however, that the mean of the *a*-parameter estimates for Form X differs from the mean for Form Y. These two means would

Table 6.7 Common-item parameter estimates rescaled using mean/sigma method's *A* and *B* with all common items (Excluding Item 27)

Item	Form X				Form Y			
	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$	<i>p</i> -value	$\hat{a}$	$\hat{b}$	$\hat{c}$
3	.7025	.3870	−1.3394	.2087	.7543	.4416	−1.3349	.1559
6	.7412	.4965	−1.5118	.2038	.7946	.5730	−1.3210	.1913
9	.5816	.6414	−.4793	.1600	.6852	.5987	−.7098	.1177
12	.5574	.5640	−.4447	.1240	.6076	.6041	−.3539	.0818
15	.4770	.9089	.6260	.2986	.5532	.9902	.5320	.3024
18	.4825	.8224	−.2749	.0535	.5587	.8081	−.1156	.0648
21	.3404	.2958	2.1735	.1489	.3655	.4140	2.5538	.2410
24	.4063	1.2396	.7002	.2453	.4774	1.3554	.5811	.2243
30	.1820	.5969	2.1304	.0853	.2231	.6336	1.8960	.0794
33	.1826	1.1972	1.3253	.0789	.2634	1.1347	1.0790	.0630
36	.1530	1.1048	2.0349	.1075	.1950	.9255	2.1337	.1259
$\hat{\mu}$	.4370	.7504	.4491	.1559	.4980	.7708	.4491	.1498
$\hat{\sigma}$	.1961	.3116	1.2935	.0728	.2018	.2858	1.2935	.0768

Fig. 6.3 Estimated test characteristic curves for common items



have been the same if the mean/mean method was used. How would the means and standard deviations of the parameter estimates compare if a characteristic curve method was used? All of these statistics would likely differ from Form X to Form Y. These results illustrate that the different methods of scaling using parameter estimates can produce different results, which in turn would affect the equating.

Test characteristic curves for the common items after the common-item parameter estimates were placed on the same θ -scale using the mean/sigma method are shown in Fig. 6.3. The Form X curve is the test characteristic curve for the 11 common items

(excluding item 27) estimated on the examinees who took Form X. The Form Y curve is the test characteristic curve for these same items estimated on the examinees who took Form Y. In general, the test characteristic curves appear to be similar. However, if the Stocking and Lord method had been used, then these test characteristic curves likely would have been even closer, because the Stocking and Lord procedure finds the A - and B -constants that minimize the difference between these characteristic curves. However, if the Stocking and Lord method had been used, then the means and standard deviations of both the a -parameter and the b -parameter estimates for the common items would have differed from Form X to Form Y.

Even after transformation to a common scale, however, the common items have different parameter estimates on Form X than they do on Form Y. These differences must be due to error in estimating the item parameters or failure of the IRT model to hold, because the items are identical on the two forms. McKinley (1988) described various methods for dealing with different parameter estimates.

The rescaled Form X item parameter estimates for all of the items are shown in Table 6.8. The same transformation that is used for the common items on Form X is also used for the other items.

6.8.2 IRT True Score Equating

The rescaled item parameter estimates then are used to estimate the true score equating function; this process is illustrated in Table 6.9 and Fig. 6.4. Figure 6.4 presents the test characteristic curves for Form X and Form Y, and Table 6.9 presents the conversion table. The equating was conducted using the PIE computer software described in Appendix B. Suppose, for example, interest focuses on finding the Form Y equivalent of a Form X score of 25. First, find the θ that is associated with a true score of 25. In Fig. 6.4, begin at a vertical axis value of 25 and go over to the Form X curve. Going down to the horizontal axis, the score of 25 is associated with a θ of approximately 1.1. With greater precision, from Table 6.9, this θ is 1.1022. This tabled value was found using the Newton-Raphson procedure that was described earlier. Next, find the Form Y true score that is associated with a θ of 1.1022. Graphically, this Form Y score is approximately 26.4. With greater precision, from Table 6.9, this true score is 26.3874. These procedures are repeated with each of the Form X integer scores, and the resulting equivalents are plotted in Fig. 6.5.

The arrows in this figure illustrate that a Form X score of 25 corresponds to a Form Y score of 26.4 (26.3874 with greater precision). Based on this graph, Form Y is easier than Form X, except at the lower scores, because the curve for true score equating is higher than the line for identity equating at all but the low scores.

In Table 6.9 θ equivalents are not given for very low Form X scores or for a Form X score of 36. The sum of the c -parameter estimates on Form X equals 6.5271, so that true score equivalents for Form X integer scores at or below a score of 6 are undefined. Kolen's (1981) ad hoc method was used to find the Form Y equivalents for these scores.

Table 6.8 Form X item parameter estimates rescaled with the mean/sigma method's A and B using all common items except item 27

Item	Form X				Form Y			
	p -value	$\hat{a}$	$\hat{b}$	$\hat{c}$	p -value	$\hat{a}$	$\hat{b}$	$\hat{c}$
1	.8440	.4673	−2.6165	.1751	.8527	.8704	−1.4507	.1576
2	.6669	.6709	−1.0683	.1165	.6161	.4628	−.4070	.1094
3	.7025	.3870	−1.3394	.2087	.7543	.4416	−1.3349	.1559
4	.5405	1.2280	.0641	.2826	.7145	.5448	−.9017	.1381
5	.6723	.8282	−.7018	.2625	.8295	.6200	−1.4865	.2114
6	.7412	.4965	−1.5118	.2038	.7946	.5730	−1.3210	.1913
7	.5895	.7316	.0304	.3224	.6351	1.1752	.0691	.2947
8	.6475	.9731	−.6572	.2209	.6094	.4450	.2324	.2723
9	.5816	.6414	−.4793	.1600	.6852	.5987	−.7098	.1177
10	.5296	.7797	.6882	.3648	.6644	.8479	−.4253	.1445
11	.4825	.8156	.3446	.2399	.7439	1.0320	−.8184	.0936
12	.5574	.5640	−.4447	.1240	.6076	.6041	−.3539	.0818
13	.5411	1.0479	−.0141	.2535	.5685	.8297	−.0191	.1283
14	.4051	.8921	.4228	.1569	.6094	.7252	−.3155	.0854
15	.4770	.9089	.6260	.2986	.5532	.9902	.5320	.3024
16	.5139	.7817	.2130	.2521	.5092	.7749	.5394	.2179
17	.5175	.7598	.0989	.2273	.4786	.5942	.8987	.2299
18	.4825	.8224	−.2749	.0535	.5587	.8081	−.1156	.0648
19	.4909	.5580	−.0511	.1201	.6265	.9640	−.1948	.1633
20	.4081	.8976	.6109	.2036	.4908	.7836	.3506	.1299
21	.3404	.2958	2.1735	.1489	.3655	.4140	2.5538	.2410
22	.4299	.7169	.7425	.2332	.5905	.7618	−.1591	.1137
23	.3839	.9473	.1809	.0644	.5092	1.1959	.5056	.2397
24	.4063	1.2396	.7002	.2453	.4774	1.3554	.5811	.2243
25	.3706	.4368	1.1176	.1427	.4976	1.1869	.6229	.2577
26	.3077	.7917	.7639	.0879	.5055	1.0296	.3898	.1856
27	.2956	1.5995	1.1495	.1992	.3771	1.0417	.9392	.1651
28	.2612	1.2792	1.2708	.1642	.3851	1.2055	1.1350	.2323
29	.2727	.8217	1.3120	.1431	.3894	.9697	.6976	.1070
30	.1820	.5969	2.1304	.0853	.2231	.6336	1.8960	.0794
31	.3059	1.0757	1.7020	.2443	.3166	1.0822	1.3864	.1855
32	.2146	.7285	1.5115	.0865	.3356	1.0195	.9197	.1027
33	.1826	1.1972	1.3253	.0789	.2634	1.1347	1.0790	.0630
34	.1814	.4939	3.5801	.1399	.1760	1.1948	1.8411	.0999
35	.1288	.7871	3.1654	.1090	.1424	1.1961	2.0297	.0832
36	.1530	1.1048	2.0349	.1075	.1950	.9255	2.1337	.1259

Note Common-item numbers and parameter estimates are in **boldface** type

6.8.3 IRT Observed Score Equating

Estimates of the distribution of θ are needed to conduct observed score equating in this example. The posterior distributions of θ that were estimated are presented in

Table 6.9 Form Y equivalents of form X scores using IRT estimated true score equating

Form X Score	θ -Equivalent	Form Y Equivalent
0		.0000
1		.8890
2		1.7760
3		2.6641
4		3.5521
5		4.4401
6		5.3282
7	−4.3361	6.1340
8	−2.7701	7.1859
9	−2.0633	8.3950
10	−1.6072	9.6217
11	−1.2682	10.8256
12	−.9951	12.0002
13	−.7633	13.1495
14	−.5593	14.2803
15	−.3747	15.3995
16	−.2043	16.5135
17	−.0440	17.6271
18	.1088	18.7429
19	.2562	19.8612
20	.3998	20.9793
21	.5409	22.0926
22	.6805	23.1950
23	.8197	24.2806
24	.9598	25.3452
25	1.1022	26.3874
26	1.2490	27.4088
27	1.4031	28.4138
28	1.5681	29.4083
29	1.7491	30.3977
30	1.9533	31.3844
31	2.1916	32.3637
32	2.4824	33.3179
33	2.8604	34.2096
34	3.3992	34.9799
35	4.3214	35.5756
36		36.0000

Table 6.10. As was noted earlier, BILOG-MG treats the posterior distribution as a discrete distribution on a finite number (10 in this example) of points. For Form X, the posterior distribution of θ needs to be converted to the ability scale of the group that took Form Y. Because the distribution is discrete, the scale conversion can be

Fig. 6.4 Estimated test characteristic curves for Form X and Form Y

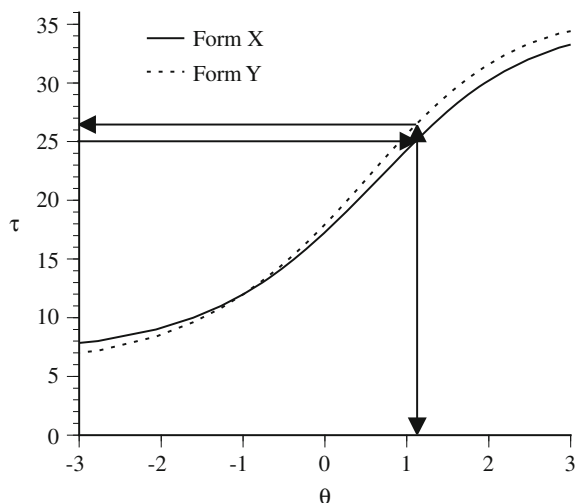
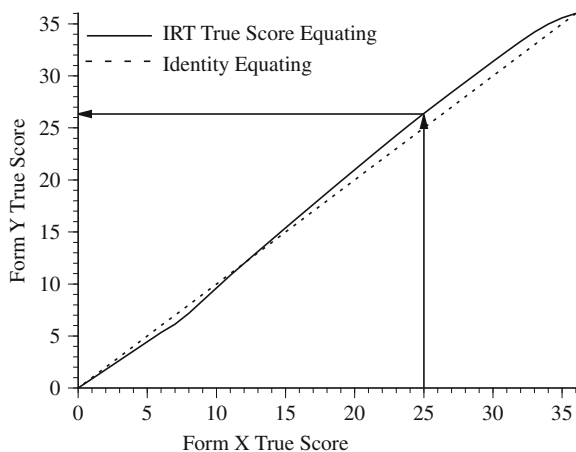


Fig. 6.5 Estimated Form Y true score equivalents of Form X true scores using IRT true score equating



accomplished by using Eq. (6.2) linearly to transform the θ -values using the A - and B -constants that were estimated earlier using the mean/sigma methods. For example, to transform the first tabled θ -value using the constants from the mean/sigma method, take $1.1761(-4.0000) - .5042 = -5.2086$, which is the tabled value. The discrete densities (ψ) do not need to be transformed.

To continue the equating process, the number-correct observed score distributions need to be estimated for the synthetic group. To simplify the presentation, the synthetic group is chosen to be the group taking Form X, so that $w_1 = 1$. In this case, estimates of $f_1(x)$ and $g_1(y)$ are needed.

The distribution of Form X number-correct scores for Group 1 can be estimated directly from the data. However, Eq. (6.27) can be used to obtain a smoothed estimate

Table 6.10 Distributions of θ estimated using BILOG

Group Taking Form X		Group Taking Form X Converted to Form Y Scale		Group Taking Form Y	
θ_I	$\hat{\psi}_1(\theta_I)$	θ_J	$\hat{\psi}_1(\theta_J)$	θ_J	$\hat{\psi}_2(\theta_J)$
−4.0000	.000101	−5.2086	.000101	−4.0000	.000117
−3.1110	.002760	−4.1630	.002760	−3.1110	.003242
−2.2220	.030210	−3.1175	.030210	−2.2220	.034490
−1.3330	.142000	−2.0720	.142000	−1.3330	.147100
−.4444	.314900	−1.0269	.314900	−.4444	.314800
.4444	.315800	.0184	.315800	.4444	.311000
1.3330	.154200	1.0635	.154200	1.3330	.152600
2.2220	.035960	2.1090	.035960	2.2220	.034060
3.1110	.003925	3.1546	.003925	3.1110	.002510
4.0000	.000186	4.2001	.000186	4.0000	.000112

of the distribution of $f_1(x)$ by using (a) the item parameter estimates for Form X converted to the Form Y scale shown in Table 6.8 and (b) the distribution of θ for the group taking Form X converted to the Form Y scale shown in Table 6.10. (In Table 6.10, the distribution of θ is approximated using 10 points to make it easier to display the distribution in the present example. However, the distribution of θ can be more accurately represented by 20 or even 40 points.)

The distribution of Form Y number-correct scores in Group 1 is not observed directly. To estimate this distribution use (a) the item parameter estimates for Form Y shown in Table 6.8 and (b) the distribution of θ for the group taking Form X converted to the Form Y scale shown in Table 6.10.

The distributions estimated using the IRT model are shown in Table 6.11 along with the equipercents that are obtained using these distributions. The equipercents were calculated using the PIE computer software described in Appendix B. (These smoothed distributions are still somewhat irregular, which might be due to the use of only 10 quadrature points. For example, modes are present at Form X scores of 11 and 25 and at Form Y scores of 11, 17, and 26.) Moments for these distributions are shown in Table 6.12, where the moments labeled “Actual” are those that came from the data without any IRT estimation. These moments were presented in Chaps. 4 and 5. The moments in the next section of Table 6.12 are for the distributions estimated using the IRT model. For example, the mean of 15.8177 for Group 1 on Form X is the mean of the distribution for Group 1 on Form X shown in the second column of Table 6.11. The Group 1 Form X moments from the two sources are quite similar. The actual mean, without any IRT estimation, was 15.8205, whereas the mean for the estimate of the distribution using the IRT model was 15.8177. Similarly, the moments for Group 2 Form Y from the two sources are similar. Similar results can be achieved for both IRT true and IRT observed score equating in this example using the C computer code described by Brennan et al. (2009, pp. 257–284).

Table 6.11 IRT observed score results using $w_1 = 1$

Score	$\hat{f}_1(x)$	$\hat{g}_1(y)$	$\hat{e}_Y(x)$
0	.0000	.0000	−.3429
1	.0001	.0002	.6178
2	.0005	.0011	1.5800
3	.0018	.0034	2.5457
4	.0050	.0081	3.5182
5	.0110	.0155	4.5021
6	.0201	.0248	5.5042
7	.0315	.0349	6.5309
8	.0437	.0446	7.5848
9	.0548	.0527	8.6604
10	.0626	.0595	9.7464
11	.0660	.0606	10.8345
12	.0651	.0589	11.9282
13	.0615	.0545	13.0431
14	.0579	.0501	14.1945
15	.0560	.0480	15.3672
16	.0555	.0488	16.5109
17	.0541	.0505	17.5953
18	.0498	.0502	18.6416
19	.0424	.0459	19.6766
20	.0338	.0379	20.7364
21	.0271	.0290	21.8756
22	.0240	.0221	23.1020
23	.0245	.0195	24.2897
24	.0261	.0209	25.3624
25	.0262	.0242	26.3651
26	.0233	.0264	27.3440
27	.0182	.0251	28.3226
28	.0132	.0205	29.3203
29	.0102	.0147	30.3521
30	.0092	.0106	31.3787
31	.0087	.0093	32.3473
32	.0072	.0092	33.2818
33	.0049	.0083	34.2001
34	.0027	.0060	35.0759
35	.0012	.0035	35.8527
36	.0003	.0014	36.3904

Because $w_1 = 1$, the moments for Group 1 are the only ones needed. In Group 1, for example, Form X is $16.1753 - 15.8177 = .3576$ points more difficult than Form Y.

The bottom portion of Table 6.12 shows the moments of converted scores for Group 1 examinees for IRT true score, IRT observed score, and frequency estimation

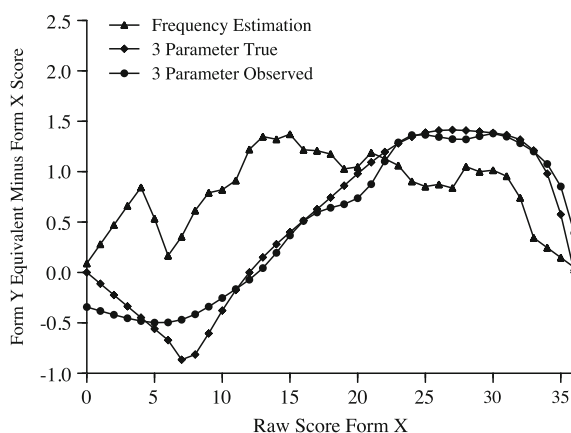
Table 6.12 Moments for equating Form X and Form Y

Group	Score	$\hat{\mu}$	$\hat{\sigma}$	$\hat{sk}$	$\hat{ku}$
Actual					
1	X	15.8205	6.5278	.5799	2.7217
2	Y	18.6728	6.8784	.2051	2.3028
Estimated using IRT observed score methods					
1	X	15.8177	6.5248	.5841	2.7235
1	Y	16.1753	7.1238	.5374	2.5750
2	X	18.0311	6.3583	.2843	2.4038
2	Y	18.6659	6.8788	.2270	2.3056
Group 1 Form X converted to Form Y scale using IRT true, IRT observed, and frequency estimation methods					
1	$\hat{\tau}_Y(x)$	16.1784	7.2038	.4956	2.5194
1	$\hat{e}_Y(x)$ IRT	16.1794	7.1122	.5423	2.5761
1	$\hat{e}_Y(x)$ Freq. Est.	16.8329	6.6017	.4622	2.6229

(from Chap. 5) equating. For example, the mean of the Form X scores converted to the Form Y scale using IRT true score equating is 16.1784; using IRT observed score equating the mean is 16.1794. The mean for frequency estimation equating is 16.8329, which was given in Table 5.10. The moments of converted scores are very similar for the two IRT methods, although the moments differ noticeably from those for frequency estimation. Note that frequency estimation included item 27 as a common item, whereas item 27 was not included as a common item for the IRT equating. This difference, and the different statistical assumptions made for frequency estimation compared to the IRT methods, likely contributed to the differences in moments that were observed.

The conversions are plotted in Fig. 6.6. In this plot, the relationship for both IRT methods differs noticeably from the frequency estimation relationship. This difference is likely a result of the very different statistical assumptions used in frequency estimation as compared to IRT. Also, the true and observed score methods relationships are similar over most of the score range. The largest differences occur around the sum of the c -parameter estimates and at the very high scores, which are near the regions of the score scale where true scores are undefined. This figure illustrates that if interest is in accurately estimating equivalents at very high scores or near the sum of the c -parameter estimates, such as when a passing score is at a point in one of these score scale regions, then distinctions between the IRT true and observed score methods need to be considered.

Fig. 6.6 Estimated equating relationships for IRT true and IRT observed score equating



6.8.4 Rasch Equating

The fit of the Rasch model to these data might not be good because these multiple-choice items are possibly subject to the effects of guessing, and the items on these forms are not built to be equally discriminating. Still, these data can be used to examine equating with the Rasch model. As was described earlier in this chapter, the Rasch model can be viewed as a special case of the three-parameter logistic model, where $D = 1.0$, all $a_j = 1$, and all $c_j = 0$.

An earlier version of BILOG-MG (BILOG 3, Mislevy and Bock 1990) was used to estimate the item parameters and posterior distributions of θ using the Rasch model. After being placed on a common scale, the Rasch item difficulty parameter estimates are shown in Table 6.13. The item difficulty estimates for the common items (after scaling) are shown in Fig. 6.7. There appear to be no outliers.

Rasch true score and observed score (with $w_1 = 1$) equating results are shown in Table 6.14, and moments are shown in Table 6.15. The equating relationships for the Rasch and three-parameter model are plotted in Fig. 6.8.

Overall, the Rasch results appear to differ from the three-parameter model results shown earlier. The Rasch observed score and true score results differ slightly at the lower scores.

These results demonstrate that Rasch observed score equating and Rasch true score equating methods are distinct. Even though Rasch true score equating is typically used in practice, Rasch observed score equating also should be considered, especially when interest is in ensuring comparability of observed score distributions. Issues in choosing among results when conducting equating in practice are discussed in Chap. 8. Because the Rasch model has relatively modest sample size requirements, this model might be considered when the sample size is small.

Table 6.13 Rasch item difficulty estimates

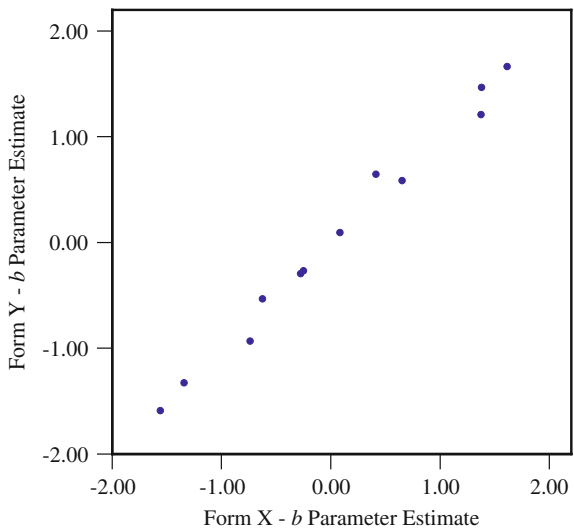
Item	Form X	Form Y
1	-2.2593	-2.0388
2	-1.1559	-.5748
3	-1.3429	-1.3275
4	-.5455	-1.0935
5	-1.1838	-1.8460
6	-1.5596	-1.5901
7	-.7757	-.6703
8	-1.0582	-.5412
9	-.7384	-.9317
10	-.4947	-.8215
11	-.2756	-1.2651
12	-.6246	-.5325
13	-.5484	-.3414
14	.0903	-.5417
15	-.2502	-.2675
16	-.4217	-.0569
17	-.4386	.0893
18	-.2757	-.2943
19	-.3150	-.6273
20	.0757	.0306
21	.4129	.6461
22	-.0285	-.4484
23	.1936	-.0570
24	.0844	.0948
25	.2594	-.0015
26	.5861	-.0396
27	.6525	.5864
28	.8508	.5463
29	.7831	.5246
30	1.3792	1.4673
31	.5958	.9051
32	1.1458	.8025
33	1.3750	1.2106
34	1.3835	1.8085
35	1.8361	2.0944
36	1.6137	1.6644

Note Common-item numbers and parameter estimates are in boldface type

6.9 Using IRT Calibrated Item Pools and Other Designs

A *calibrated item pool* (Lord 1980; Vale 1986) is a group of items that have item parameter estimates which have all been placed on the same θ -scale. One potential benefit of using IRT is that calibrated item pools can be constructed, and the item

Fig. 6.7 Plots of Rasch difficulty estimates on Form X versus Form Y



parameter estimates can be used directly in equating. Equating designs that use calibrated item pools often allow for greater flexibility in constructing test forms than the other designs that have been described previously. In this section, the development of IRT calibrated item pools, and how they are used in equating, are described.

6.9.1 Common-Item Equating to a Calibrated Pool

Consider the following simplified example of how an IRT calibrated item pool might evolve. Form Y is constructed and then administered. A transformation is developed to convert scores on Form Y to scale scores, and the item parameters for Form Y also are estimated. So far, equating has not been considered, because there is only a single form.

Form X_1 is constructed next. Form X_1 contains some new items and some items in common with Form Y. Form X_1 is administered to a new group of examinees, and the item parameters are estimated for the new form. Form X_1 can be equated to Form Y using the common-item equating procedures described earlier in this chapter. Along with a conversion table for Form X_1 scores, this common-item equating procedure results in item parameter estimates for Form X_1 which are on the ability scale that was established with Form Y. Actually, there is now a calibrated pool of items, some of which were in Form Y only, some of which were in Form X_1 only, and some of which were in both forms. Refer to Table 6.8. The item parameter estimates in this table are all on the same θ -scale. The items in this table could be considered to be an IRT calibrated item pool.

The use of an IRT calibrated item pool makes possible the use of an equating design that is similar to the common-item nonequivalent groups design. However,

Table 6.14 Rasch true and observed score equating results

x	$\hat{t}_Y(x)$	$\hat{e}_Y(x)$
0	.0000	.6995
1	1.0780	1.2612
2	2.1550	2.3202
3	3.2280	3.3782
4	4.2953	4.4318
5	5.3563	5.4739
6	6.4107	6.5024
7	7.4586	7.5207
8	8.5002	8.5419
9	9.5358	9.5670
10	10.5655	10.5914
11	11.5896	11.6098
12	12.6083	12.6206
13	13.6218	13.6257
14	14.6302	14.6275
15	15.6336	15.6280
16	16.6322	16.6274
17	17.6260	17.6239
18	18.6150	18.6153
19	19.5994	19.6010
20	20.5793	20.5809
21	21.5546	21.5560
22	22.5257	22.5272
23	23.4925	23.4956
24	24.4554	24.4628
25	25.4147	25.4269
26	26.3707	26.3891
27	27.3241	27.3486
28	28.2754	28.3047
29	29.2255	29.2587
30	30.1757	30.2137
31	31.1275	31.1711
32	32.0827	32.1302
33	33.0439	33.0914
34	34.0142	34.0572
35	34.9978	35.0302
36	36.0000	36.0113

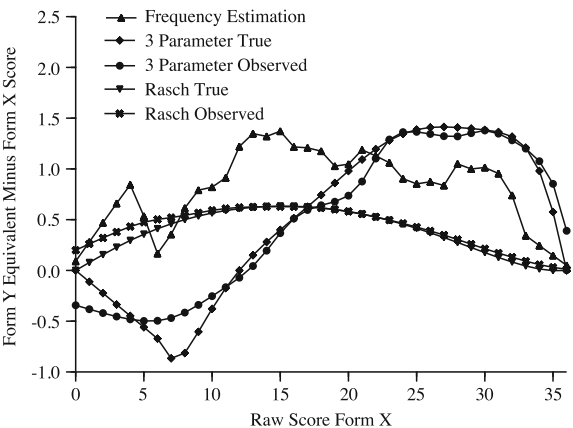
in this new design, the common items are drawn from the pool rather than from a single old form. This new design is referred to here as *common-item equating to a calibrated pool*.

To describe this design, suppose that another new form, Form X_2 , is constructed. This form consists of a set of common items from the IRT calibrated item pool and

Table 6.15 Moments for equating Form X and Form Y using Rasch equating

Group	Score	$\hat{\mu}$	$\hat{\sigma}$	$\hat{sk}$	$\hat{ku}$
Actual					
1	X	15.8205	6.5278	.5799	2.7217
2	Y	18.6728	6.8784	.2051	2.3028
Estimated using Rasch observed score methods					
1	X	15.8307	6.4805	.3658	2.5974
1	Y	16.3808	6.4388	.3107	2.5542
2	X	18.1342	6.9291	.1328	2.3458
2	Y	18.6553	6.8406	.0810	2.3438
Group 1 Form X converted to Form Y scale using Rasch					
True, Rasch observed, and frequency estimation methods					
1	$\hat{\gamma}(x)$	16.3554	6.4685	.5212	2.6521
1	$\hat{e}_Y(x)$ Rasch	16.3830	6.4266	.3156	2.5559
1	$\hat{e}_Y(x)$ Freq. Est.	16.8329	6.6017	.4622	2.6229

Fig. 6.8 Estimated equating relationships for three-parameter and Rasch true and observed score equating



some new items. Assume that Form X_2 is administered to a group of examinees. Procedures described earlier can be used to transform the IRT scale that results from estimating Form X_2 item parameters to the scale that was established for the pool. To implement these procedures, the item parameter estimates from the calibrated pool for the common items are considered to be on Scale J , and the item parameter estimates from the calibration of Form X_2 are considered to be on Scale I .

After the new form item parameter estimates are transformed to the θ -scale for the calibrated pool, IRT estimated true score or observed score equating could be conducted. Estimated true score equating for Form X_2 could be implemented as follows. First, find the θ that corresponds to each Form X_2 integer number-correct score. Finding these θ values requires an iterative procedure as described earlier. Second, find the Form Y true score equivalent of each of the θ -values. Following this step results in a true score equating of Form X_2 to Form Y. Use the Form Y scale

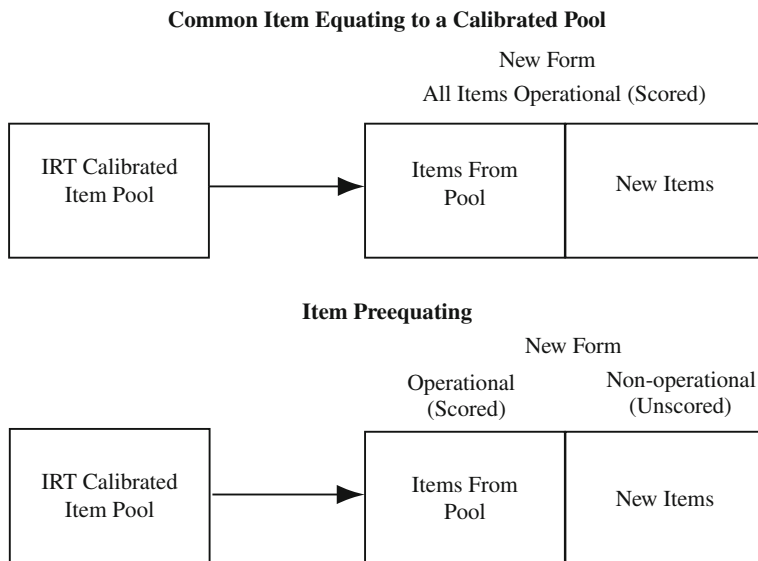


Fig. 6.9 Equating designs that use an IRT calibrated item pool

score transformation to convert the Form X_2 integer number-correct scores to scale scores. These procedures are very similar to what is done in common-item equating, with the major difference being that the common items are taken from a calibrated item pool rather than from a single previously equated form.

After the equating is completed, the new Form X_2 items have item parameter estimates on the θ -scale that was established for the pool. These new items can be added to the IRT calibrated item pool. In this way, the pool can be continually expanded. The common-item sets for new forms are constructed from a continually increasing IRT calibrated item pool. A diagram representing common-item equating to a calibrated pool is presented in the top portion of Fig. 6.9.

Many practical issues affect the implementation of IRT calibrated item pools in practice. For example, items might be removed from a pool because their content becomes dated or for test security purposes. Also, when items are used more than once, procedures need to be considered for updating the parameter estimates that are associated with each item in the pool. (For example, two sets of item parameter estimates exist for each common item in Table 6.8.) These are among the issues that are considered when using item pools in a testing program.

Common-item equating to a calibrated pool is more flexible than the common-item nonequivalent groups design, because it allows the common-item set to be chosen from many previous test forms rather than from a single test form. The effects of violations of IRT assumptions need to be considered, however, when using this design. For example, the position of items can affect their performance. For this

reason, the position of each common item on the new form should be as close as possible to its position on the form in which it appeared previously.

Also, real tests are typically not strictly unidimensional. To guard against multidimensionality causing problems with equating, as with traditional equating, the set of common items should be built to the same content specifications, proportionally, as the total test. In this way, the violations of assumptions might affect the common items in the same way that they affect the total scores. Also, a large enough number of common items should be chosen to represent fully the content of the total test.

IRT might be the only procedure that could be used when equating using common-item equating with a calibrated item pool. What if the IRT assumptions are severely violated? Then adequate equating might be impossible with this design. For this reason, if common-item equating to a calibrated item pool is being considered for use, the common-item nonequivalent groups design should be used for a few administrations. The results for the IRT method and traditional methods could be compared and the effects of multidimensionality could be assessed. Switching to common-item equating with a calibrated item pool should be done only if no problems are found with that procedure.

6.9.2 Item Preequating

The use of IRT calibrated item pools also makes an *item preequating design* possible. The goal of item preequating is to be able to produce raw-to-scale score conversion tables *before* a form is administered intact. If a conversion table is produced before the test form is administered, then scores can be reported to examinees without the need to wait for equating to be conducted. Item preequating is possible if the items that contribute to examinees' scores have been previously administered and calibrated.

Consider the following example of how an item preequating design might evolve. Form Y is developed. Form Y contains *operational items*, which are items that contribute to examinees' scores. Form Y also contains *nonoperational items*, which are items that do not contribute to examinees' scores. A conversion of Form Y number-correct scores on the operational items to scale scores is constructed. (The scale could be defined either before or after administration of Form Y.) Form Y is administered and item parameters of the operational and nonoperational items are estimated. At this point, the IRT calibrated item pool consists of the operational and the nonoperational Form Y items that have parameter estimates on the same IRT scale. So far, equating has not been considered, because there is only a single form.

The operational component of a new form, Form X₁, could be constructed from this calibrated pool of items. If so, the operational component of Form X₁ would consist of some combination of Form Y operational and Form Y nonoperational items. Because the operational items in Form X₁ already have estimated item parameters, a conversion table could be constructed for the operational component of Form X₁ before Form X₁ was ever administered intact. That is, the operational portion of Form X₁ could be "preequated." Form X₁ also would contain nonoperational items,

which would be newly written items that were not yet part of the item pool. After Form X_1 was administered, the item parameters for all Form X_1 items (operational and nonoperational) could be estimated. The operational Form X_1 items then could be used as the set of common items for transforming the item parameter estimates for the nonoperational items to the θ -scale that was established with Form Y. These nonoperational Form X_1 items then would be added to the calibrated item pool. The operational portion of subsequent test forms would be constructed from the calibrated pool. The nonoperational portion of subsequent test forms would consist of new items, and would be used to expand the item pool continually.

A diagram representing the item preequating design is presented in the bottom portion of Fig. 6.9. The item preequating design and common-item designs differ as to whether or not scores on the new items contribute to examinee scores. These designs also differ in whether or not conversion tables can be produced before the new form is administered.

A variety of issues need to be considered when using item preequating in practice. Suppose it is found that the answer key for an operational item needs to be modified (e.g., an item needs to be double-keyed) after the test is administered. Then the preequating would need to be modified.

In addition, to ensure that items will behave the same on each administration, items should appear in contexts and positions when they appear operationally that are similar to those used when they appear nonoperationally. Although item preequating has been found to produce acceptable results (Bejar and Wingersky 1982), problems can occur when the nonoperational items are presented in a separate section. For example, Eignor (1985), Eignor and Stocking (1986), and Stocking and Eignor (1986) conducted a series of studies that suggested problems with item preequating if it were used with the SAT. Kolen and Harris (1990) found similar problems with item preequating if it was used with the ACT tests. Context effects and multidimensionality were suggested as reasons for these problems. On the other hand, Quenette et al. (2006) obtain reasonably stable IRT item preequating with the ASVAB. In situations where the context and positions of items cannot be fixed from one testing to the next, formal studies need to be conducted to make sure that the resulting data will produce fairly robust parameter estimates and equated scores.

The use of item preequating can cause difficulties in estimating the item parameters for the nonoperational items. For example, assume that a test is not strictly unidimensional. In this case, IRT estimation procedures will estimate some composite of multidimensional abilities. The appropriate composite for a test will be the composite for forms that are all built to the test specifications. Estimates that are based only on considering the operational items would estimate this composite. Consider a situation in which the nonoperational items do not represent well the test content specifications. What would happen if the nonoperational item parameters were estimated in the same run using IRT computer software as the operational items? The composite that is estimated might differ from the composite that would result if only the operational items were used. The use of a different composite might lead to bias in the estimation of item parameters. Although it might be possible to handle estimation problems in practice, the scenario just described suggests that

estimation can be quite complicated when estimating parameters for nonoperational items. The problems just described can affect parameter estimates whenever nonoperational items are used in tests that are equated using IRT methods under any of the equating designs described in this book, such as whenever items are tried out (pretested) for inclusion in future forms.

On the surface, item preequating seems straightforward. However, its implementation can be quite complicated. Context effects and dimensionality issues need to be carefully considered, or misleading results will be likely.

6.9.3 Other Designs

Many variations on designs for equating using IRT exist. For example, new forms might consist of items in common with a pool, new operational items, and nonoperational items. Such pools can be used to produce computer administered and computer adaptive tests (see Chap. 8 for a brief discussion). Glas and Béguin (2011) provide another example of a complex design. No attempt will be made here to enumerate all of these variations. However, context effects and dimensionality issues that arise with each variation need to be carefully considered when using item pools in operational testing programs.

6.10 Equating with Polytomous IRT

For the IRT models discussed so far, it has been assumed the items are scored dichotomously. When items are scored in more than two categories, dichotomous models are not appropriate, and polytomous IRT models can be used. In this section, the focus is on equating with polytomously scored items in which the responses are ordered. Typically, the responses are ordered so that responses to higher categories are associated with better performance on the item, although it is possible for the ordering to be in the other direction. Kim et al. (2010) reviewed methods for equating using polytomous IRT models and associated research.

One situation where polytomous IRT models can be used is when writing samples are collected from students and are scored holistically by raters, say, on a scale from 1 to 5. In addition, sometimes mixed-format tests contain a mixture of polytomously and dichotomously scored items, such as on a test that contains both multiple-choice and constructed-response test questions.

Another situation occurs when multiple items are associated with a common stimulus block, as often occurs in reading comprehension tests. Because there could be some dependency among items associated with a particular stimulus, violations of local independence might make the use of dichotomous IRT questionable. To address this problem, items associated with a common stimulus could be scored as a block, with scores on the block of items ranging from 0 to the number of items in the block.

For analysis purposes, a block of items could be treated as a single polytomous item. For example, a 15-dichotomous item reading test containing 3 passages with 5 items each could be treated as a test with 3 polytomous items (blocks), with scores on each polytomous item ranging from 0 to 5. These blocks of items are sometimes referred to as testlets (Thissen et al. 1989). Keller et al. (2003) discussed issues associated with potential loss of precision when testlets are scored in this way. Lee et al. (2001) compared equating based on polytomous and dichotomous IRT models in the testlet situation and found that the polytomous models produced more accurate equating. Wainer et al. (2007) and DeMars (2012) investigated the use of IRT models designed for testlets.

Many of the same considerations associated with dichotomous models come into play when IRT equating is conducted with polytomous models. With polytomous models, scales can be linked using generalizations of the item characteristic curve linking methods, and generalizations of IRT true and IRT observed score equating methods can be used to equate total scores.

In the polytomous models considered here, each item is scored in two or more ordered categories. As with dichotomous models, examinee ability is described by a single variable, θ , defined so that $-\infty < \theta < \infty$. The category response curve for each category of an item relates the probability of earning the category score to examinee ability. The category response curve for category k of item j is symbolized as $p_{jk}(\theta)$, which represents the probability that an examinee of ability θ receives a score in category k . For example, if 10 % of the examinees with ability 1.5 can be expected to earn a score in category 3 on item 1, then $p_{13}(\theta = 1.5) = .10$. Each category of the item has a category response curve.

As with dichotomous models, local independence for polytomous IRT models means that after taking into account examinee ability, examinee responses to the items are statistically independent. So, for example, if examinees with $\theta = 1.5$ have a .1 probability of earning a score in category 3 for item 1 *and* a .4 probability of earning a score in category 4 for item 2, their probability of earning a score in category 3 for item 1 and a score in category 4 for item 2 equals $.04 = .1(.4)$.

6.10.1 Polytomous IRT Models for Ordered Responses

Various polytomous IRT models have been developed that can be used to model items that are scored polytomously using ordered categories. These include models suggested by Samejima's (1969) and Bock (1972), and more recently described by Samejima (1997) and Bock (1997). Samejima designated the categories of each item with consecutive integers beginning with 0. Bock designated categories with consecutive integers beginning with 1. In this section, Bock's designation is used, even in describing Samejima's model, for consistency sake. However, as described later in this section, a scoring function is also introduced, that might differ from the category designator. This scoring function is used to accommodate the scores as used by Samejima as well as other item scoring schemes.

Samejima's Graded Response Model

Although originally developed as a normal ogive model, Samejima's (1969) graded response model also has been presented in the logistic form that is considered here. The graded response model directly models the cumulative category response function. The cumulative category response function for category k of item j is the probability of earning a score at or above category k on that item. For this model, the probability that persons of ability equal to that of person i will earn a score on item j at or above category k can be expressed as

$$\begin{aligned} p_{ijk}^*(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}) &= 1, & k &= 1, \\ p_{ijk}^*(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}) &= \frac{\exp[Da_j(\theta_i - b_{jk})]}{1 + \exp[Da_j(\theta_i - b_{jk})]}, & k &= 2, \dots, m_j. \end{aligned} \quad (6.29)$$

For the first category, the cumulative category response function is 1, because the probability is 1 that any examinee, regardless of their θ , will earn a score at or above the first category. In this equation, D is a scaling factor (usually 1.7 so that the logistic is similar to the cumulative normal) and a_j is the item slope parameter. The item has m_j categories, and b_{jk} are item difficulty parameters for categories 2 through m_j . The first category does not have a difficulty parameter. For categories 2 through m_j , the expression is essentially the item characteristic function for the two-parameter logistic model.

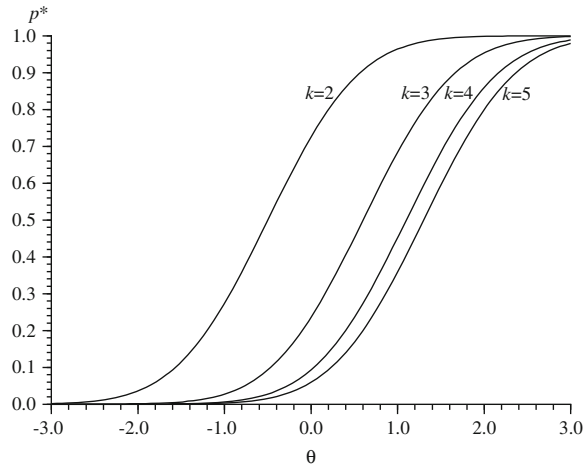
The category response function is calculated by taking the difference between the cumulative category response functions as follows:

$$\begin{aligned} p_{ijk}(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}) &= p_{ijk}^*(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}) \\ &\quad - p_{ij(k+1)}^*(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}), \quad k = 1, \dots, m_j - 1, \\ p_{ijk}(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}) &= p_{ijk}^*(\theta_i; a_j, b_{j_2}, \dots, b_{j_{m_j}}), \quad k = m_j. \end{aligned} \quad (6.30)$$

As an example, consider a 5-category item with parameters $a = 1.2$, $b_2 = -.5$, $b_3 = .6$, $b_4 = 1.1$, $b_5 = 1.3$. Using Eq. (6.29), the cumulative category response function for this item at $\theta = 1.0$ can be shown to be .964, .684, .452, and .359 for categories 2 through 5. Then from Eq. (6.30), the category response function is .036 = (1 - .964) for the first category, 0.28 = (.964 - .684) for the second category, 0.232 = (.684 - .452) for the third category, and 0.093 = (.452 - .359) for the fourth category. As is always the case, the category response function for the last category is equal to the cumulative category response function for the last category. For this item, this probability is .359.

The cumulative category response functions for this item, which represent the probability of earning a score at or above a particular category, are graphed in Fig. 6.10. Note that the cumulative category response functions are parallel, which is always the case for Samejima's graded response model. Also note that the curves

Fig. 6.10 Cumulative category response functions for a graded response model item



are farther apart when the differences between adjacent b -parameters are large. For example, the difference between the b -parameters for category 2 and 3 is 1.1 units $[-.6 - (-.5)]$, which is the largest difference between adjacent b -parameters.

The category response functions for this item are graphed in Fig. 6.11. The category response function for the first category decreases as θ increases. The category response function for the last category increases as θ increases. The first and last categories can be expected to have this pattern with polytomous models for items with ordered categories, as long as higher category designations tend to be associated with higher θ . The intermediate categories have category response functions that all begin with probability near 0, increase to their maximum probability, and then decrease to a probability near zero. Intermediate categories for polytomous models for items with ordered response items typically have curves of this form. The highest point for an intermediate curve is greater when the differences between adjacent b -parameters are large. Thus, for example, the curve for the second category is the highest among the intermediate curves in Fig. 6.11.

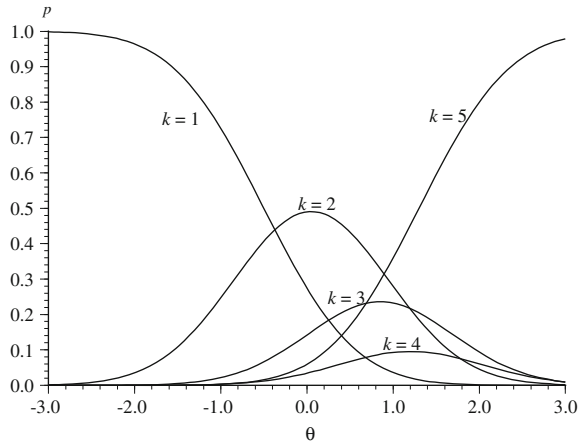
Bock's Nominal Model

Bock's (1972) nominal model can be used to model polytomous items that have ordered or unordered categories. The category response function for this model is

$$p_{ijk}(\theta_i; a_{j1}, a_{j2}, \dots, a_{jm_j}, c_{j1}, c_{j2}, \dots, c_{jm_j}) = \frac{\exp(a_{jk}\theta_i + c_{jk})}{\sum_{h=1}^{m_j} \exp(a_{jh}\theta_i + c_{jh})}. \quad (6.31)$$

Each category for an item has a slope parameter, a_{jk} , and an intercept parameter, c_{jk} . This model is very general. It can be shown that if the slope parameters, a_{jk} ,

Fig. 6.11 Category response functions for a graded response model item



increase from one category to the next, such that $a_{j1} < a_{j2} < \dots < a_{jm_j}$, then this model can be used to represent items with ordered categories (Bock 1997; Samejima 1972; Wainer et al. 1991). Thissen et al. (1995) described how to fit this model when responses are ordered using polynomial contrasts on the slope parameters.

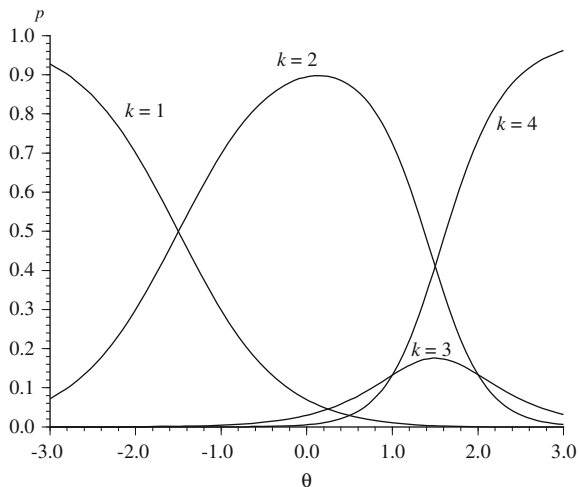
As an example, consider an item with four categories with a_{jk} parameters of 1.7, 3.4, 5.1, 6.8, and c_{jk} parameters of 0.0, 2.55, -0.85 , -2.55 . Note that the a_{jk} parameters increase as category increases, consistent with this item's having ordered categories. For this item, for example, the reader can verify that the probability of an examinee with $\theta = 1$ earning a score in category 1 is 0.010; and in categories 2–4, the probabilities are 0.725, 0.132, and 0.132, respectively. The category response function for this item is shown in Fig. 6.12. As can be seen in this figure, the general shapes of the functions for the first and last categories are similar to those for the Samejima graded response model item discussed previously. The general shapes of intermediate curves are also similar for the two models.

Various other models can be viewed as being special cases of the nominal categories model. Muraki's (1992, 1997) generalized partial credit model is one of these. In this model,

$$p_{ijk}(\theta_i; a_j^*, b_j, d_{j1}, d_{j2}, \dots, d_{jm_j}) = \frac{\exp \left[\sum_{h=1}^k D a_j^* (\theta_i - b_j + d_{jh}) \right]}{\sum_{g=1}^{m_j} \exp \left[\sum_{h=1}^g D a_j^* (\theta_i - b_j + d_{jh}) \right]}. \quad (6.32)$$

In this equation, D is a scaling constant (typically 1.7), item parameters are the discrimination parameter a_j^* and the difficulty parameter, b_j . There are also difficulty parameters for each category, $d_{j1}, d_{j2}, \dots, d_{jm_j}$. This model is overparameterized as

Fig. 6.12 Category response functions for a generalized partial credit model item



stated, and sometimes the parameters are set as follows: $b_j = 0$ and $d_{j1} = 0$. An alternative parameterization is sometimes used in which a category difficulty parameter is used that is the difference between b_j and d_{jk} . In this section, the parameterization shown in Eq. (6.32) is used.

The form of this equation, with the single summation in the numerator and double summation in the denominator, is more complicated than the other IRT models discussed so far. As an example of how this equation would be implemented for a three category item the numerator is

$$\begin{aligned} &\exp[Da_j^*(\theta_i - b_j + d_{j1})], \text{ for category 1,} \\ &\exp[Da_j^*(\theta_i - b_j + d_{j1}) + Da_j^*(\theta_i - b_j + d_{j2})], \text{ for category 2, and} \\ &\exp[Da_j^*(\theta_i - b_j + d_{j1}) + Da_j^*(\theta_i - b_j + d_{j2}) + Da_j^*(\theta_i - b_j + d_{j3})], \text{ for category 3.} \end{aligned}$$

The denominator is the sum of these three numerators.

As an example, consider an item with four categories with $D = 1.7$, $a_j^* = 1$, $b_j = 0$, and d_{jk} of 0, 1.5, -2, and -1 for the four categories. For this item, the reader can verify that the probability of an examinee with $\theta = 1$ earning a score in category 1 is 0.010, and in categories 2–4, respectively, the probabilities are 0.725, 0.132, and 0.132. Note that these four probabilities are the same as the probabilities for Bock's nominal model example discussed earlier. In addition, the category response function for this item is the same as that for Bock's nominal model item shown in Fig. 6.12. Because Muraki's generalized partial credit model is a special case of Bock's nominal model, there are Bock's nominal model parameters that correspond to the generalized partial credit model parameters. This Muraki's generalized partial credit model example was purposefully chosen to have the same model parameters as Bock's nominal model example.

The relationship between the parameters for the two models is expressed as follows:

$$\begin{aligned}
 a_{jk} &= Dka_j^*, \text{ and} \\
 c_{jk} &= -Dka_j^*b_j + Da_j^* \sum_{h=1}^k d_{jh}.
 \end{aligned}
 \tag{6.33}$$

If Muraki's generalized partial credit model parameters for the example are substituted in this equation, Bock's nominal model parameters in the earlier example are obtained.

The relationship between the parameters of these two models and the appearance of the category response functions is much less clear than the relationships for dichotomous items or for Samejima's graded response models. For the purposes of this chapter, it is primarily important to note that these models can be used with test items that have ordered categories. These models provide descriptions of the category response functions for these types of items. In addition, there are other models discussed by Bock (1997) and Muraki (1997) that can be viewed as special cases of Muraki's generalized partial credit model.

6.10.2 Scoring Function, Item Response Function, and Test Characteristic Curve

Often, total scores are used with test items that are polytomously scored. A scoring function is used to associate the scores with the categories. Let W_{jk} refer to the integer score associated with category k . Often a scoring function of $W_{jk} = k$ is used. In this case, a response associated with the first category earns a score of 1, a response associated with the second category earns a score of 2, and so forth. Another scoring function that is often used is $W_{jk} = k - 1$. For this function, a response associated with the first category earns a 0, a response associated with the second category earns a score of 1, and so forth.

Based on a scoring function, the minimum and maximum scores on test Form X can be calculated as

$$\begin{aligned}
 \min_X &= \sum_{j:X} W_{j1}, \\
 \max_X &= \sum_{j:X} W_{j m_j}.
 \end{aligned}
 \tag{6.34}$$

Thus, to obtain the minimum, the minimum scores for items are summed over items on Form X. To obtain the maximum, the maximum scores for items are summed. Note that if all items are scored using a minimum for each item as zero, then the minimum score is zero. If the minimum score for each item is 1, then the minimum score for Form X equals the number of items.

The item response function relates total score on an item to θ . This function is expressed as

$$\tau_j(\theta_i) = \sum_{k=1}^{m_j} W_{jk} p_{ijk}(\theta_i), \quad (6.35)$$

where $p_{ijk}(\theta_i)$ is the category response function for item j for a polytomous IRT model.

For polytomous IRT models, the test characteristic curve for Form X is calculated as

$$\tau_X(\theta_i) = \sum_{j:X} \tau_j(\theta_i). \quad (6.36)$$

Similar to dichotomous IRT models, the test characteristic curve relates IRT ability to true total scores.

6.10.3 *Parameter Estimation and Scale Transformation with Polytomous IRT Models*

Item and ability parameters for Samejima's graded response model and Bock's nominal model can be estimated using the computer software MULTILOG (Thissen et al. 2003). PARSCALE (Muraki and Bock 2003) can be used to estimate parameters for Samejima's graded response model and for the generalized partial credit model. ICL (Hanson 2002) can be used for the generalized partial credit model. These programs can also estimate item parameters on mixed format tests that contain multiple item types.

As with dichotomous IRT models, for the random groups or single group designs, as long as the item parameters are estimated using the same scaling conventions (e.g., mean ability of 0 and standard deviation of ability 1), then the estimates from separate runs on Form X and Form Y are on the same scale. For the single group design, the item and ability parameters for Samejima's graded response model and Bock's nominal model can be estimated with MULTILOG in a single computer run.

For the common item non-equivalent groups design, ICL, Multilog, and PARSCALE can be used to concurrently estimate parameters for the old and new form. Alternatively, when test forms are administered to nonequivalent groups, scale transformation methods can be used with the polytomous IRT models that are analogous to those for dichotomous IRT models. Methods using moments of item parameter estimates are given first followed by characteristic curve methods. Many of these methods were provided in greater detail by Kim and Kolen (2005). Although not considered in detail in this section, Kim (2010) generalized the Ogasawara (2001a) scale transformation method referenced earlier in this chapter that takes into account error in estimating item parameters.

6.10.3.1 Mean/Sigma and Mean/Mean Methods

Mean/mean and mean/sigma methods were suggested by Cohen and Kim (1998) for the graded response model. For the mean/sigma method, the mean and standard deviation of the b -parameter estimates are found over all items and all categories. The mean and standard deviation of the b -parameter estimates for the common items are calculated separately for the old form and new form calibration. For example, if there are 5 common items with 4 score categories each, then there are 15 b -parameter estimates in the common item set (5 items times 4 – 1 = 3 b -parameter estimates per item in each calibration). The resulting means and standard deviations are substituted for the parameters in Eqs. (6.8a) and (6.9a) to obtain the slope and intercept of the transformation equation. Equations (6.2), (6.3), and (6.4) are used to transform the θ -, a -, and b -parameters. The mean/mean method uses the mean of the b -parameter estimates as calculated for the mean/sigma method as well as the mean of the a -parameter estimates over the common items. Equations (6.8b) and (6.9a) are used to obtain the slope and intercept of the transformation function.

A similar process can be followed for scale linking using the mean/mean and mean/sigma method with Muraki's generalized partial credit model shown in Eq. (6.32). For the mean/sigma method, the mean and standard deviation of the estimates of $b_j - d_{jh}$ are found over all items and categories for each calibration. These standard deviations are substituted for the standard deviations in Eq. (6.8a) to calculate the slope and the means substituted in Eq. (6.9a) to find the intercept of the transformation equation. The mean/mean method uses the means of the a^* parameter estimates and the means of the estimates of $b_j - d_{jh}$. Equations (6.8b) and (6.9a) are used to obtain the slope and intercept of the transformation function. The a^* parameter estimates are transformed using Eq. (6.3). The b parameter estimates are transformed using Eq. (6.4). The d parameter estimates are transformed by multiplying them by the slope computed using Eq. (6.8a). The θ -estimates are transformed using Eq. (6.2). Masters (1984) described linking procedures for the partial credit model, a special case of the generalized partial credit model, which involve only adding a constant for this model.

6.10.3.2 Test Characteristic Curve Methods

Test characteristic curve methods can be used with polytomous IRT models. For polytomous IRT models, it is necessary to establish the criteria over categories within item as well as over items. The Haebara difference for the graded response model is

$$Hdiff(\theta_i) = \sum_{j:V} \sum_{k:j} \left[p_{ijk}(\theta_{ji}; \hat{a}_{Jj}, \hat{b}_{Jj2}, \dots, \hat{b}_{Jjk}, \dots, \hat{b}_{Jjm_j}) - p_{ijk} \left(\theta_{ji}; \frac{\hat{a}_{ij}}{A}, A\hat{b}_{Ij2} + B, \dots, A\hat{b}_{Ijk} + B, \dots, A\hat{b}_{Ijm_j} + B \right) \right]^2. \quad (6.37)$$

The first summation is over items and the second is over categories within item. Thus this function is the sum of squared differences between category response curves over all categories and items. $Hcrit$ is found by substituting Eq. (6.37) in Eq. (6.12). This criterion is minimized by summing over examinees as discussed with the dichotomous model.

The Stocking and Lord difference for the graded response model is

$$SLdiff(\theta_i) = \left[\sum_{j:V} \sum_{k:j} W_{jk} p_{ijk}(\theta_{Ji}; \hat{a}_{Jj}, \hat{b}_{Jj2}, \dots, \hat{b}_{Jjk}, \dots, \hat{b}_{Jjm_j}) - \sum_{j:V} \sum_{k:j} W_{jk} p_{ijk} \left(\theta_{Ji}; \frac{\hat{a}_{Jj}}{A}, A\hat{b}_{Jj2} + B, \dots, A\hat{b}_{Jjk} + B, \dots, A\hat{b}_{Jjm_j} + B \right) \right]^2. \quad (6.38)$$

Recall that the Stocking and Lord approach was based on the squared difference between the test characteristic curves expressed on the two scales. Referring to Eqs. (6.35) and (6.36), it can be seen that this equation is the squared difference between test characteristic curves. Note that the scoring function (W_{jk}) is used in $SLdiff$ but not in $Hdiff$. $SLcrit$ is found by substituting Eq. (6.38) in Eq. (6.15). This criterion is minimized by summing over examinees as discussed with the dichotomous model.

Baker (1992) developed a Stocking and Lord related method for the graded response model. Baker's (1993a) EQUATE 2.0 program can be used with the Stocking and Lord approach, using a fixed set of abilities to cumulate over abilities. Other ways of cumulating over ability for the Stocking and Lord method are implemented in POLYST listed in Appendix B. Baker (1993b, pp. 249, 250) described a procedure for minimizing $Hcrit$ for this model, which is also implemented in POLYST. Also, see Brennan et al. (2009, pp. 223–256) for C code that can be used with these models.

$Hdiff$ and $SLdiff$ are defined similarly for the generalized partial credit model. With this model, though, it is necessary to also transform the d -parameter estimates.

For Bock's nominal model,

$$Hdiff(\theta_i) = \sum_{j:V} \sum_{k:j} \left[p_{ijk} \left(\theta_{Ji}; \hat{a}_{Jj1}, \dots, \hat{a}_{Jjk}, \dots, \hat{a}_{Jjm_j}, \dots, \hat{a}_{Jm_j} \right) - p_{ijk} \left(\theta_{Ji}; \frac{\hat{a}_{Jj1}}{A}, \dots, \frac{\hat{a}_{Jjk}}{A}, \dots, \frac{\hat{a}_{Jjm_j}}{A}, \dots, \frac{\hat{a}_{Jm_j}}{A}, \dots, \hat{c}_{Jj1} - \frac{B}{A}\hat{a}_{Jj1}, \dots, \hat{c}_{Jjk} - \frac{B}{A}\hat{a}_{Jjk}, \dots, \hat{c}_{Jjm_j} - \frac{B}{A}\hat{a}_{Jjm_j}, \dots, \hat{c}_{Jm_j} - \frac{B}{A}\hat{a}_{Jm_j} \right) \right]^2. \quad (6.39)$$

$Hcrit$ is found by substituting Eq. (6.39) in Eq. (6.12). This criterion is minimized by summing over examinees as discussed with the dichotomous model. Baker (1993b) described this method summing over equally spaced points and it is implemented in his EQUATE 2.0 computer software (Baker 1993a). Kim and Hanson (2002)

provided a correction to one of Baker's (1993b) equations. Generalizations of these methods are implemented in POLYST and in Brennan et al. (2009, pp. 223–256).

The Stocking and Lord procedure can be implemented for this model in situations when modeling graded response data. For Bock's nominal model,

$$SLdiff(\theta_i) = \left[\begin{array}{c} \sum_{j:V} \sum_{k:j} W_{jk} P_{ijk} \left(\theta_{ji}; \hat{a}_{Jj1}, \dots, \hat{a}_{Jjk}, \dots, \hat{a}_{Jjm_j}, \right) - \\ \sum_{j:V} \sum_{k:j} W_{jk} P_{ijk} \left(\theta_{ji}; \frac{\hat{a}_{Ij1}}{A}, \dots, \frac{\hat{a}_{Ijk}}{A}, \dots, A\hat{a}_{Ijm_j}, \right. \\ \left. \hat{c}_{Ij1} - \frac{B}{A}\hat{a}_{Ij1}, \dots, \hat{c}_{Ijk} - \frac{B}{A}\hat{a}_{Ijk}, \dots, \right. \\ \left. \hat{c}_{Ijm_j} - \frac{B}{A}\hat{a}_{Ijm_j} \right) \end{array} \right]^2. \quad (6.40)$$

As pointed out by Baker (1993b), this procedure is not appropriate when items are nominally scored, because in this case scoring weights would not typically be available. Thus, this procedure can be used only with items scored in ordered categories. This method is implemented in POLYST and in Brennan et al. (2009, pp. 223–256).

In addition to the models considered here, scale linking methods have been developed for a testlet IRT model (Li et al. 2005), a continuous response IRT model (Shojima 2003), an unfolding IRT model (Koenig and Roberts 2007), and a non-parametric IRT model (Xu et al. 2011). Kim (2006) developed a method for using information on distractors to improve IRT linking with Bock's nominal response model.

Research on Scale Linking in Polytomous IRT

In a simulation study, Cohen and Kim (1998) compared the mean/mean, mean/sigma, weighted mean/sigma, Stocking and Lord (1983) extension, and an extension of Divgi's (1985) method that Kim and Cohen (1995) developed for linking scales under the graded response model. They concluded that the methods produced similar results. Baker (1997) studied the empirical sampling distributions of the linking coefficients under the graded response model. Kim and Cohen (2002) compared linking using the Stocking and Lord method and concurrent calibration for data that were simulated to fit the graded response model. They found that concurrent calibration was slightly more accurate. Clearly, more research on linking methods and comparisons between linking methods and concurrent calibration for polytomous IRT models is needed.

Kim and Lee (2004) applied scaling linking methods to mixed-format tests that contained both dichotomously and polytomously scored items in a simulation study. They found the MULTILOG and PARSCALE produced results that were similarly accurate. They also found that characteristic curve methods produced more accurate results than the mean/mean and mean/sigma methods.

6.10.4 True Score Equating

Using Eq.(6.36) to calculate IRT true scores, the true score equating process described for dichotomous models in conjunction with Eq.(6.19) is used, except that typically there is no lower asymptote parameter in the polytomous models.

6.10.5 Observed Score Equating

IRT observed score equating for polytomous IRT models is very similar to that for dichotomous IRT models. The major difference is that the distribution of observed score given IRT ability is modeled using a compound multinomial distribution, which is a generalization of the compound binomial distribution described earlier. A recursion formula that was described by Thissen et al. (1995) can be used to perform the calculations.

Define $f_1(x = W_{11}|\theta_i) = p_{i11}(\theta_i)$ as the probability of earning a score in the first category of item 1, $f_1(x = W_{12}|\theta_i) = p_{i12}(\theta_i)$ as the probability of earning a score in the second category of item 1, and so forth up to the last category of item 1. Then for $r > 1$, the recursion formula for finding the probability of earning score x after the r -th item added is,

$$f_r(x|\theta_i) = \sum_{k=1}^{m_j} f_{r-1}(x - W_{jk})p_{ijk}(\theta_i) \text{ for } x \text{ between } \min_r \text{ and } \max_r, \quad (6.41)$$

where $\min_r$ and $\max_r$ are the minimum and maximum scores after adding the r -th item. Note that when $x - W_{jk} < \min_{r-1}$ or $x - W_{jk} > \max_{r-1}$, then $f_{r-1}(x - W_{jk}) = 0$, by definition.

An example using the recursive formula is given in Table 6.16. This example is for a three-item test, where each item has a scoring function that consists of consecutive integers beginning with 1. The first and second items have four categories each. The third item has three categories. In this table, the i subscript for ability is dropped to simplify the table. To use the recursion formula, it is important to identify the maximum and minimum score after each new item is added. For the first item ($r = 1$) the minimum score is 1 and the maximum is 4. When the second item is added ($r = 2$), the minimum is 2 and the maximum is 8. After the third item is added ($r = 3$), the minimum is 3 and the maximum is 11. In Table 6.16, a zero is displayed whenever $x - W_{jk}$ is less than the minimum score or greater than the maximum score.

A computational example that goes along with the recursive example in Table 6.16 is given in Table 6.17. Assume that $\theta = 1$. For this example, the first item is Bock's nominal model item used as an example earlier. The probabilities for this item as well as the other two items, conditional on $\theta = 1$, are given at the bottom of the table. The outcome of applying the recursion formula in this example is a distribution of total

Table 6.16 Polytomous IRT recursive formula example

r	x	$f_r(x)$				
1	1	$f_1(1)$	$= p_{11}$			
	2	$f_1(2)$	$= p_{12}$			
	3	$f_1(3)$	$= p_{13}$			
	4	$f_1(4)$	$= p_{14}$			
2	2	$f_2(2)$	$= f_1(1)p_{21}$	$+0$	$+0$	$+0$
	3	$f_2(3)$	$= f_1(2)p_{21}$	$+f_1(1)p_{22}$	$+0$	$+0$
	4	$f_2(4)$	$= f_1(3)p_{21}$	$+f_1(2)p_{22}$	$+f_1(1)p_{23}$	$+0$
	5	$f_2(5)$	$= f_1(4)p_{21}$	$+f_1(3)p_{22}$	$+f_1(2)p_{23}$	$+f_1(1)p_{24}$
	6	$f_2(6)$	$= 0$	$+f_1(4)p_{22}$	$+f_1(3)p_{23}$	$+f_1(2)p_{24}$
	7	$f_2(7)$	$= 0$	$+0$	$+f_1(4)p_{23}$	$+f_1(3)p_{24}$
	8	$f_2(8)$	$= 0$	$+0$	$+0$	$+f_1(4)p_{24}$
3	3	$f_3(3)$	$= f_2(2)p_{31}$	$+0$	$+0$	
	4	$f_3(4)$	$= f_2(3)p_{31}$	$+f_2(2)p_{32}$	$+0$	
	5	$f_3(5)$	$= f_2(4)p_{31}$	$+f_2(3)p_{32}$	$f_2(2)p_{33}$	
	6	$f_3(6)$	$= f_2(5)p_{31}$	$+f_2(4)p_{32}$	$f_2(3)p_{33}$	
	7	$f_3(7)$	$= f_2(6)p_{31}$	$+f_2(5)p_{32}$	$f_2(4)p_{33}$	
	8	$f_3(8)$	$= f_2(7)p_{31}$	$+f_2(6)p_{32}$	$f_2(5)p_{33}$	
	9	$f_3(9)$	$= f_2(8)p_{31}$	$+f_2(7)p_{32}$	$f_2(6)p_{33}$	
	10	$f_3(10)$	$= 0$	$+f_2(8)p_{32}$	$f_2(7)p_{33}$	
	11	$f_3(11)$	$= 0$	$+0$	$f_2(8)p_{33}$	

scores on this three-item test for examinees with $\theta = 1$. Note that the total scores range from 3 to 11.

For IRT observed score equating, the recursion formula, along with a quadrature distribution for θ , is used to find the marginal distribution for Form X using Eq. (6.26) and implemented using Eqs. (6.26) or (6.27). Similar procedures are used for Form Y. These distributions are then equated using equipercentile methods in the same way that the scores were equated in observed score equating with dichotomous IRT; the main difference is that the total scores range between the minimum and maximum score rather than between 0 and K_X .

6.10.6 Example Using the Graded Response Model

A new real data example is used to illustrate use of the graded response model in equating. The test in this example is Level 9 of the Maps and Diagrams of the Iowa Tests of Basic Skills (ITBS). Two forms of this test (Form L and Form K) were administered using a random groups design. Each form contains 24 items. There are 5 stimuli on each form of the test. The first two stimuli each have 3 items associated with them and the last three stimuli have 6 items associated with them. The items associated with each stimulus block were assumed to be a testlet. The testlet score

Table 6.17 Polytomous IRT recursive formula computational example

r	x	$f_r(x)$					
1	1	$f_1(1)$	=	.010			
	2	$f_1(2)$	=	.725			
	3	$f_1(3)$	=	.132			
	4	$f_1(4)$	=	.132			
2	2	$f_2(2)$	=	.010(.15)	+0	+0	+0 = .0015
	3	$f_2(3)$	=	.725(.15)	+.010(.25)	+0	+0 = .1112
	4	$f_2(4)$	=	.132(.15)	+.725(.25)	+.010(.40)	+0 = .2050
	5	$f_2(5)$	=	.132(.15)	+.132(.25)	+.725(.40)	+.010(.20) = .3448
	6	$f_2(6)$	=	0	+.132(.25)	+.132(.40)	+.725(.20) = .2308
	7	$f_2(7)$	=	0	+0	+.132(.40)	+.132(.20) = .0792
	8	$f_2(8)$	=	0	+0	+0	+.132(.20) = .0264
3	3	$f_3(3)$	=	.0015(.05)	+0	+0	= .0001
	4	$f_3(4)$	=	.1112(.05)	+.0015(.60)	+0	= .0065
	5	$f_3(5)$	=	.2050(.05)	+.1112(.60)	.0015(.35)	= .0775
	6	$f_3(6)$	=	.3448(.05)	+.2050(.60)	.1112(.35)	= .1792
	7	$f_3(7)$	=	.2308(.05)	+.3448(.60)	.2050(.35)	= .2902
	8	$f_3(8)$	=	.0792(.05)	+.2308(.60)	.3448(.35)	= .2631
	9	$f_3(9)$	=	.0264(.05)	+.0792(.60)	.2308(.35)	= .1296
	10	$f_3(10)$	=	0	+.0264(.60)	.0792(.35)	= .0436
	11	$f_3(11)$	=	0	+0	.0264(.35)	= .0092

Note $p_{11} = .01, p_{12} = .725, p_{13} = .132, p_{14} = .132, p_{21} = .15, p_{22} = .25, p_{23} = .40, p_{24} = .20, p_{31} = .05, p_{32} = .60, p_{33} = .35$

was the total number correct on that testlet. Each examinee had 5 scores, one for each testlet. The range of scores for the first two testlets was 0 to 3. The range of scores for the last three testlets was 0 to 6. The total score on the test ranged from 0 to 24. Examinee testlet scores were input into the computer software MULTILOG. Defaults were used for the analyses, with the exception that 49 equally spaced quadrature points ranging from -6 to $+6$ were used. IRT equating was conducted using the POLYEQUATE computer software that is given in Appendix B.

The item parameter estimates that were obtained in two runs of MULTILOG are given in Table 6.18. Because the random groups design was used, the groups taking the two forms are assumed equivalent, and the item parameters from the two runs assumed to be on the same scale, without transformation. As can be seen, each item has an a_j -parameter estimate and one less b_{jk} -parameter estimate than the number of score categories.

The true score equating results are given in Table 6.19. To conduct observed score equating, it was necessary to have quadrature distributions. MULTILOG does not print out the quadrature weights. To obtain the weights, the following process was used, which produces weights that are similar to the prior weights used by BILOG-MG. Begin with a set of quadrature points that are equally spaced and centered around zero. Find the density of the standard normal distribution at each point.

Table 6.18 Graded response model item parameter estimates

Form	Testlet	Item Parameter Estimates						
		a_j	b_{j2}	b_{j3}	b_{j4}	b_{j5}	b_{j6}	b_{j7}
L	1	1.197	-1.906	.103	1.713			
	2	1.029	-2.094	-.208	2.020			
	3	1.672	-2.355	-1.481	-.830	-.197	.551	1.670
	4	1.033	-2.272	-.706	.576	1.912	3.267	5.126
	5	1.048	-1.904	-.604	.567	1.683	2.944	4.346
K	1	1.407	-3.081	-1.179	.363			
	2	1.891	-1.851	-1.016	-.026			
	3	2.143	-2.476	-1.736	-1.174	-.594	.020	.961
	4	1.471	-2.286	-1.121	-.137	.795	1.717	2.840
	5	1.442	-2.043	-1.108	-.279	.519	1.312	2.475

Sum the weights over the points and then divide each weight by this sum, which standardizes the weights to sum to one. For the example, this process was followed with 49 quadrature points (rounded to one decimal place) ranging from -6 to +6. The results from the observed score equating are shown in Table 6.20. In addition, the frequency distributions that were obtained from the IRT model are displayed in Table 6.20. Moments of the actual and estimated distributions and the converted scores are shown in Table 6.21 (these moments were calculated using the actual, not the smoothed, relative frequency distributions).

In addition to the graded response model, the three-parameter logistic model (3PL) was also fit to the data. In this case, each form was analyzed as having 24 dichotomously scored items. Also, unsmoothed equipercetile equating was conducted. Only final results are provided for these equatings.

The observed and fitted frequency distributions are shown in Fig. 6.13. As can be seen, there appears to be a slight distortion in the fitted distribution for the graded response model, with the mode being a bit too high. This finding is consistent with the mean for Form K estimated using the graded response model (14.1708) being slightly too large compared to the actual mean (14.0066). Difference plots for all of the equatings that were conducted are shown in Fig. 6.14. The three-parameter logistic model true score method produced different results at the low scores than the other methods, presumably because of the pseudo-chance level parameter.

6.11 Robustness to Violations of the Unidimensionality Assumption

A unidimensionality assumption is required to use the IRT methods discussed in this chapter. Research suggests that IRT equating is fairly robust to violations of the unidimensionality assumption when equating alternate forms of a test, as long as

Table 6.19 Graded response model true score equating

Form L Score	θ Equivalent	Form K Equivalent
0		0.0000
1	-2.8734	1.2173
2	-2.4186	2.4598
3	-2.0852	3.7232
4	-1.7878	5.0624
5	-1.5063	6.4754
6	-1.2335	7.9417
7	-0.9675	9.4249
8	-0.7093	10.8485
9	-0.4582	12.1977
10	-0.2117	13.5085
11	0.0345	14.8072
12	0.2846	16.0576
13	0.5430	17.2252
14	0.8146	18.3479
15	1.1007	19.4247
16	1.3908	20.3425
17	1.6782	21.0697
18	1.9769	21.6899
19	2.3174	22.3015
20	2.7380	22.9719
21	3.2746	23.5915
22	3.9718	23.9136
23	4.8693	23.9903
24		24.0000

the violation of the unidimensionality assumption is not too severe (e.g., Bolt 1999; Camilli et al. 1995; Cook et al. 1985; De Champlain 1996; Dorans and Kingston 1985; Yen 1984).

Some investigators have suggested using multidimensional IRT models for tests that violate the unidimensionality assumption. In such cases, methods must be used to link the multidimensional IRT parameter estimates (Davey et al. 1996; Hirsch 1989; Li and Lissitz 2000; Oshima et al. 2000; Reckase 2009; Yao 2011; Yao and Boughton 2009). Methods for conducting IRT true and observed score equating for use with tests that are fit with a multidimensional model have been presented by Brossman (2010) for a situation in which the multiple-choice items on an examination are treated as measuring a different unidimensional construct than the constructed-response items on the same examination. Practical issues associated with equating mixed-format tests are discussed more fully in Chap. 8 and practical issues associated with deciding on whether to use a unidimensional or multidimensional model with mixed-format tests is discussed in more detail in Chap. 9

Table 6.20 Graded response model observed score equating

Score	Form L Relative frequency	Form K Relative frequency	Form K Equivalent of Form L raw score
0	.0032	.0015	.5120
1	.0067	.0041	1.6295
2	.0118	.0068	2.8232
3	.0191	.0102	4.0965
4	.0290	.0143	5.4718
5	.0408	.0188	6.8479
6	.0533	.0240	8.2768
7	.0649	.0296	9.6702
8	.0745	.0357	10.9918
9	.0810	.0426	12.2716
10	.0846	.0501	13.5346
11	.0863	.0572	14.7981
12	.0865	.0631	16.0148
13	.0838	.0661	17.1592
14	.0765	.0671	18.2546
15	.0643	.0693	19.3020
16	.0489	.0733	20.2601
17	.0343	.0752	21.0926
18	.0229	.0723	21.8007
19	.0144	.0657	22.3942
20	.0082	.0574	23.0970
21	.0037	.0470	23.6443
22	.0012	.0314	24.2817
23	.0002	.0134	24.4676
24	.0000	.0039	24.4981

Table 6.21 Moments for graded response model equating

	$\hat{\mu}$	$\hat{\sigma}$	$\hat{sk}$	$\hat{ku}$
Actual				
Form L	10.8047	4.3171	0.2256	2.4343
Form K	14.0066	5.0146	-0.2638	2.2285
Estimated using graded response observed score method				
Form L	10.7900	4.1695	-0.0442	2.5432
Form K	14.1708	4.9801	-0.3757	2.4903
Form L converted to Form K using various methods				
Equipercntile	14.0105	5.0046	-0.2577	2.2244
IRT Obs	14.1363	5.0362	-0.1279	2.1586
IRT True	14.0504	5.1688	-0.1735	2.1276

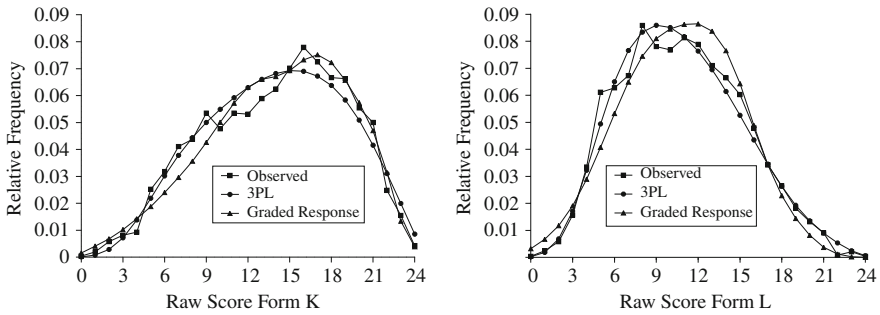
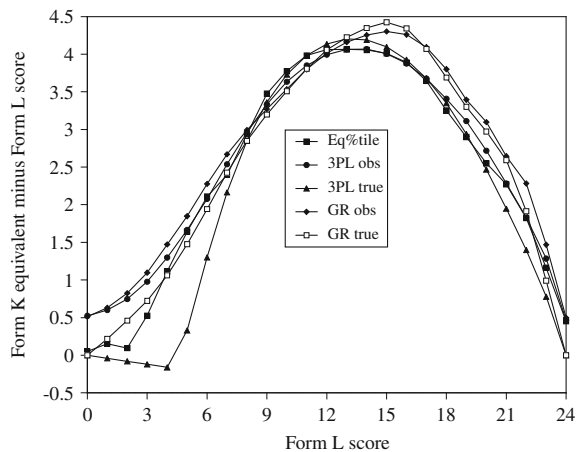


Fig. 6.13 Observed and fitted relative frequency distributions for Form K and Form L for graded response model example

Fig. 6.14 Equating relationships for graded response model example



6.12 Practical Issues and Caveat

We recommend the following when using IRT to conduct equating in practice:

1. When equating with the common item nonequivalent groups design, use both the Stocking and Lord and Haebara methods for scale transformation. In addition, concurrent calibration should be used as a check when feasible.
2. When equating number-correct scores, use both IRT true score equating and IRT observed score equating.
3. Whenever possible, conduct traditional equipercentile or linear methods on the forms that are being equated as a check.

Often all of the methods applied provide similar equating results and conversion to scale scores (where appropriate), which is reassuring. However, when the results for the different methods diverge, then a choice must be made about which results to believe. The assumptions required and the effects of poor parameter estimates need

to be considered in these cases. The issue of choosing among results in equating is discussed in more detail in Chap. 8.

Unidimensional IRT methods assume that the test forms are unidimensional and that the relationship between ability and the probability of correct response follows a specified form. These requirements are difficult to justify for many educational achievement tests, although, as indicated in the previous section, the methodology has been found to be robust to violations in many practical situations.

The IRT methods considered in detail in this chapter do not include parameters for item context effects such as item position in a test booklet. Yet, there is considerable evidence (see Chap. 8) that the difficulty of items is influenced by item context. The general approach taken in this chapter, and in this book as a whole, is to recommend that equating studies be designed to minimize the effects of violations of assumptions. In this regard, the following advice from Cook and Petersen (1987) is especially relevant:

Regardless of whether IRT true-score or conventional equating procedures are being used, common items should be selected that are a miniature of the tests to be equated and these items should remain in the same relative position when administered to the new- and old-form groups. It would also seem prudent to evaluate the differential difficulty of the common items administered to the equating samples, particularly when equating samples come from different administration dates. (p. 242)

6.13 Exercises

- 6.1 For the test in Table 6.1, find the probability of correctly answering each of the three items for examinees with ability $\theta_{Ii} = .5$.
- 6.2 For the test in Table 6.1, find the distribution of observed scores for examinees with ability $\theta_{Ii} = .5$.
- 6.3 Prove the following:
 - a. $A = (b_{Jj} - b_{Jj^*}) / (b_{Ij} - b_{Ij^*})$ from Eq. (6.6). [Hint: The proof can be done by setting up a pair of simultaneous equations for b_{Jj^*} and b_{Jj} using Eq. (6.4) and solving for A .]
 - b. $A = a_{Ij} / a_{Jj}$ from Eq. (6.6). [Hint: Use Eq. (6.3).]
 - c. $A = \sigma(b_J) / \sigma(b_I)$ in Eq. (6.8a). [Hint: Use Eq. (6.4).]
 - d. $A = \mu(a_I) / \mu(a_J)$ in Eq. (6.8b). [Hint: Use Eq. (6.3).]
- 6.4 For the test in Table 6.1, what is the value of the test characteristic curve at $\theta_{Ii} = -2.00, .5$, and 1.00 ? How about at $\theta_{Ii} = -1.50$ and 0.00 ?
- 6.5 For the hypothetical example in Table 6.3, conduct observed score equating for a population of examinees with equal numbers of examinees at three score levels: $\theta = -1, 0, 1$. [Hints: Use Eq. (6.25) to find $f(x|\theta)$ and $g(y|\theta)$ for $\theta = -1, 0$, and 1 . Then apply Eq. (6.27). Finally, do conventional equipercentile equating. *Warning:* This problem requires considerable computation.]

- 6.6 For the example in Table 6.4, provide the probabilities of earning scores 0, 1, 2, 3, and 4 for $r = 4$ assuming that the probability of correctly answering the fourth item for an examinee of ability $\theta_i = -2$ equals .4.
- 6.7 For the example in Table 6.2, calculate $Hdiff$ and $SLdiff$ for $\theta = 1$ on Scale J using the mean/sigma and mean/mean methods.
- 6.8 Why is IRT equating to a particular old form important if all items are in an IRT calibrated item pool?
- 6.9 The following are some of the steps involved in equating (assume that number-correct scoring is used and that scale scores are reported to examinees): (a) select the design for data collection and how to implement it; (b) construct, administer, and score the test; (c) estimate equating relationships; (d) construct a conversion table of raw-to-scale scores; (e) apply the conversions to examinees; and (f) report scores to examinees. At each of these steps, what would be the differences in equating a new form using the IRT methods described in Chap. 6 versus the traditional methods described in Chaps. 2, 3, 4, 5?
- 6.10 Find $p_{ijk}^*(\theta_i; a_j, b_{j2}, \dots, b_{jm_j})$ and $p_{ijk}(\theta_i; a_j, b_{j2}, \dots, b_{jm_j})$ at $\theta_i = -.5$ for a Samejima Logistic graded response model item with the following parameters: $a_j = 1.2$, $b_{j2} = -1.1$, $b_{j3} = -1.0$, $b_{j4} = .5$, $b_{j5} = .6$, and $b_{j6} = 1.0$.
- 6.11 Find $p_{ijk}(\theta_i; a_{j1}, a_{j2}, \dots, a_{jm_j}, c_{j1}, c_{j2}, \dots, c_{jm_j})$ at $\theta_i = .5$ for a Bock's nominal model item with the following parameters: $a_{j1} = .905$, $a_{j2} = .522$, $a_{j3} = -.959$, $c_{j1} = .336$, $c_{j2} = -.206$, $b_{j3} = .126$.
- 6.12 Is the item in the preceding exercise consistent with being an item with ordered categories? Why or why not?
- 6.13 Find $p_{ijk}(\theta_i; a_j^*, b_j, d_{j1}, d_{j2}, \dots, d_{jm_j})$ at $\theta_i = 1.0$ for a Muraki generalized partial credit model item with the following parameters: $a_j = 1$, $b_j = 0$, $d_{j1} = 0$, $d_{j2} = 1$, $d_{j3} = -1$.
- 6.14 For the example in Table 6.17, find the probability of earning scores of 4 through 14 if on a fourth item, the probability of earning a 1 was .3, the probability of earning a 2 was .5, and the probability of earning a 3 was .2. Use the recursive formula.
- 6.15 For the example in Table 6.17, what is the (conditional) expected score on item 1? On item 2? What is the (conditional) expected score on a two-item test consisting of the first two items? What relationship is there between these three expected scores? Why? In the terminology of the chapter, what are each of these (conditional) expected scores?
- 6.16 Show that Eq. (6.33) relates Muraki's generalized partial credit model parameters to Bock's nominal model parameters.

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